

Frontier League Single-Variable Econometric Report

Run mode: batch_all_stats

Dependent variable: WinPct

Candidate regressors analyzed: 47

Each section includes an OLS summary, diagnostic tests, and publication-quality plots.

RA9 Regression Summary

Formula: Q("Pitching_W-L%") ~ Q("Pitching_RA9")

Coefficient Table

term	coef	std_err	t_stat	p_value	ci_lower	ci_upper
Intercept	0.9459	0.0421	22.4491	0.0	0.862	1.0298
Q("Pitching_RA9")	-0.0793	0.0074	-10.7551	0.0	-0.094	-0.0646

Model Fit

	value
R-squared	0.5973
Adjusted R-squared	0.5921
F-statistic	115.6731
F-statistic p-value	0.0000
Observations	80.0000

Heteroskedasticity Tests

Test	LM Statistic	LM p-value	F Statistic	F p-value
Breusch-Pagan	3.0314	0.0817	3.0721	0.0836
White	4.0714	0.1306	2.0644	0.1339

Normality Test

JB statistic = 1.0651, JB p-value = 0.5871, Skewness = 0.0987, Kurtosis = 2.4703

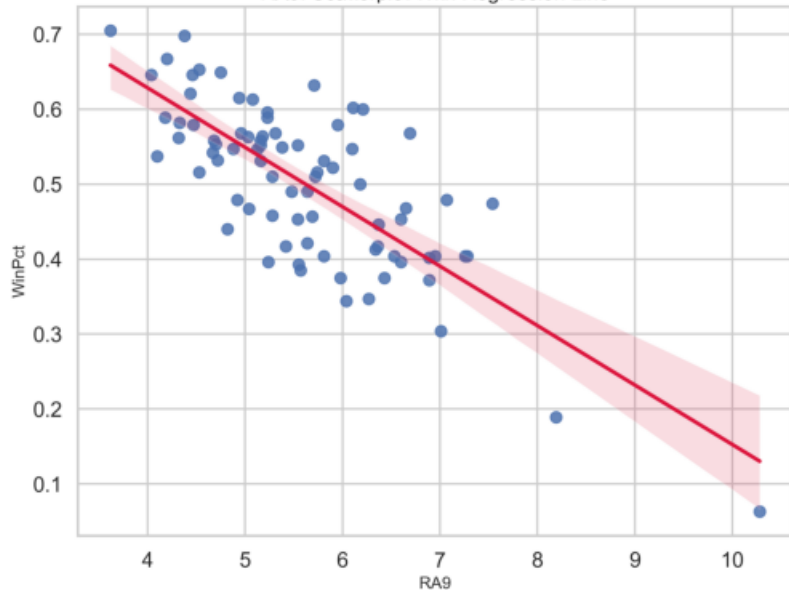
Interpretation

RA9 has a statistically significant negative relationship with WinPct and explains approximately 59.7% of the variation in team winning percentage. RA9 residuals do not exhibit significant heteroskedasticity under either the Breusch-Pagan or White tests. Residual normality is not rejected at the 5% level.

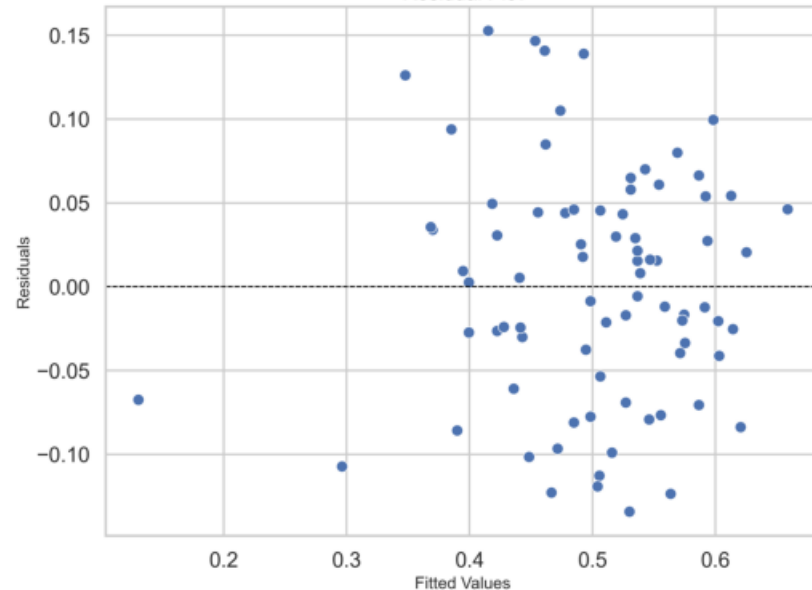
Warnings

- 4 observations exceed the leverage threshold of 0.0500.
- 4 observations exceed the Cook's distance threshold of 0.0500.

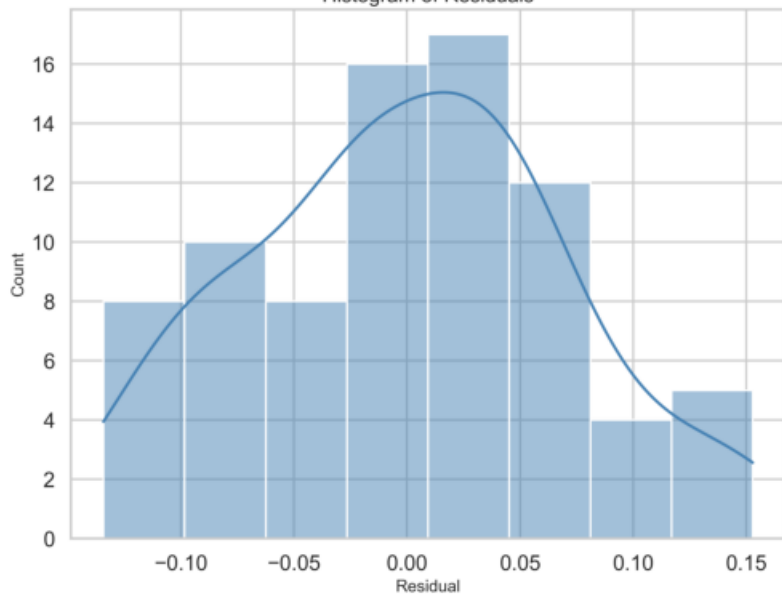
RA9: Scatterplot With Regression Line



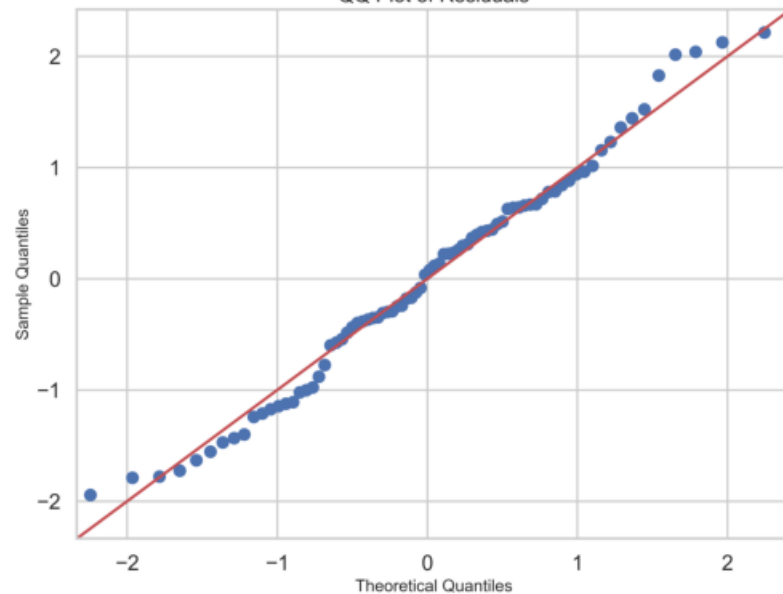
Residual Plot



Histogram of Residuals



QQ Plot of Residuals



ERA Regression Summary

Formula: Q("Pitching_W-L%") ~ Q("Pitching_ERA")

Coefficient Table

term	coef	std_err	t_stat	p_value	ci_lower	ci_upper
Intercept	0.9230	0.0423	21.8146	0.0	0.8388	1.0073
Q("Pitching_ERA")	-0.0867	0.0085	-10.1707	0.0	-0.1036	-0.0697

Model Fit

	value
R-squared	0.5701
Adjusted R-squared	0.5646
F-statistic	103.4440
F-statistic p-value	0.0000
Observations	80.0000

Heteroskedasticity Tests

Test	LM Statistic	LM p-value	F Statistic	F p-value
Breusch-Pagan	1.1553	0.2824	1.1429	0.2883
White	2.2701	0.3214	1.1244	0.3301

Normality Test

JB statistic = 1.2269, JB p-value = 0.5415, Skewness = 0.0243, Kurtosis = 2.3953

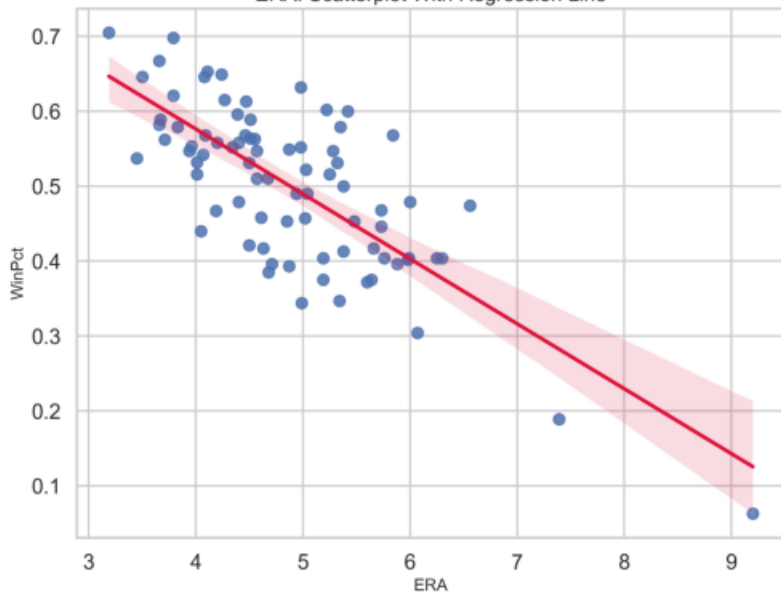
Interpretation

ERA has a statistically significant negative relationship with WinPct and explains approximately 57.0% of the variation in team winning percentage. ERA residuals do not exhibit significant heteroskedasticity under either the Breusch-Pagan or White tests. Residual normality is not rejected at the 5% level.

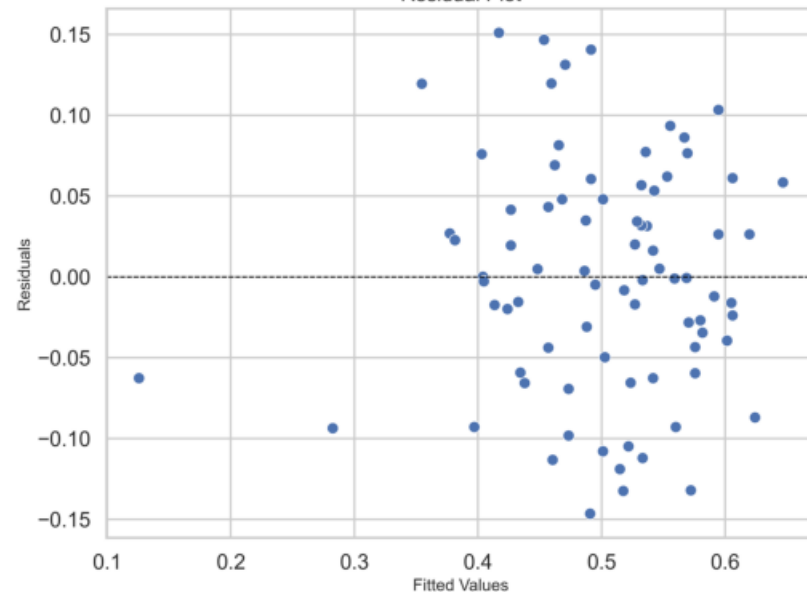
Warnings

- 4 observations exceed the leverage threshold of 0.0500.
- 4 observations exceed the Cook's distance threshold of 0.0500.

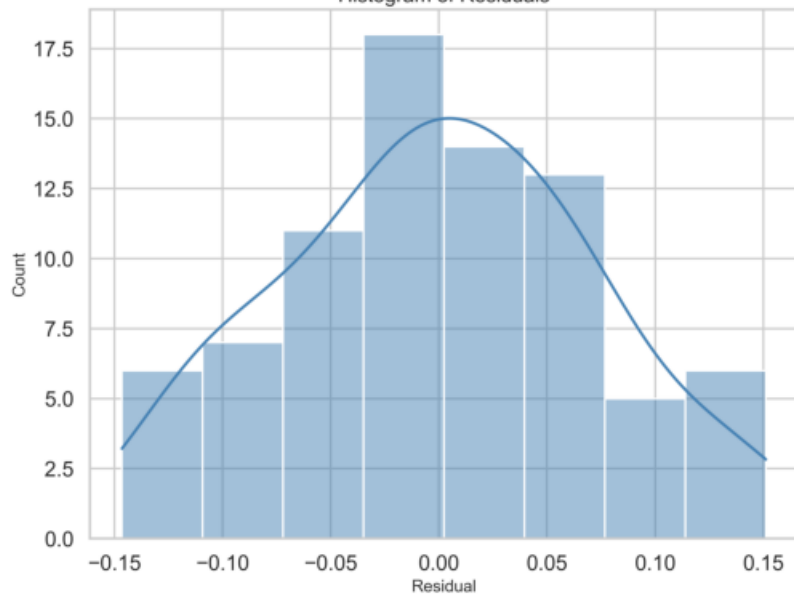
ERA: Scatterplot With Regression Line



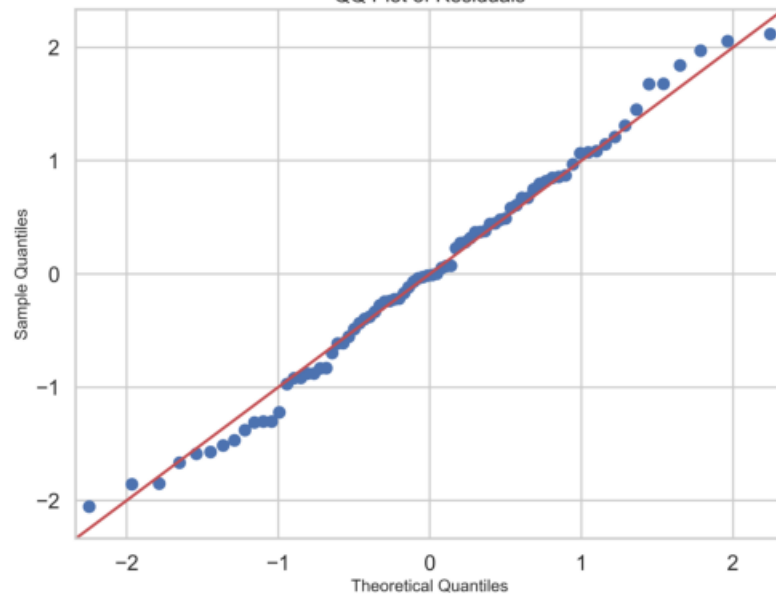
Residual Plot



Histogram of Residuals



QQ Plot of Residuals



Pitching_ER Regression Summary

Formula: Q("Pitching_W-L%") ~ Q("Pitching_ER")

Coefficient Table

term	coef	std_err	t_stat	p_value	ci_lower	ci_upper
Intercept	0.9534	0.0482	19.7907	0.0	0.8575	1.0493
Q("Pitching_ER")	-0.0010	0.0001	-9.5444	0.0	-0.0012	-0.0008

Model Fit

	value
R-squared	0.5387
Adjusted R-squared	0.5328
F-statistic	91.0959
F-statistic p-value	0.0000
Observations	80.0000

Heteroskedasticity Tests

Test	LM Statistic	LM p-value	F Statistic	F p-value
Breusch-Pagan	3.6068	0.0575	3.6826	0.0586
White	3.6099	0.1645	1.8194	0.1690

Normality Test

JB statistic = 1.2965, JB p-value = 0.5230, Skewness = 0.0178, Kurtosis = 2.3774

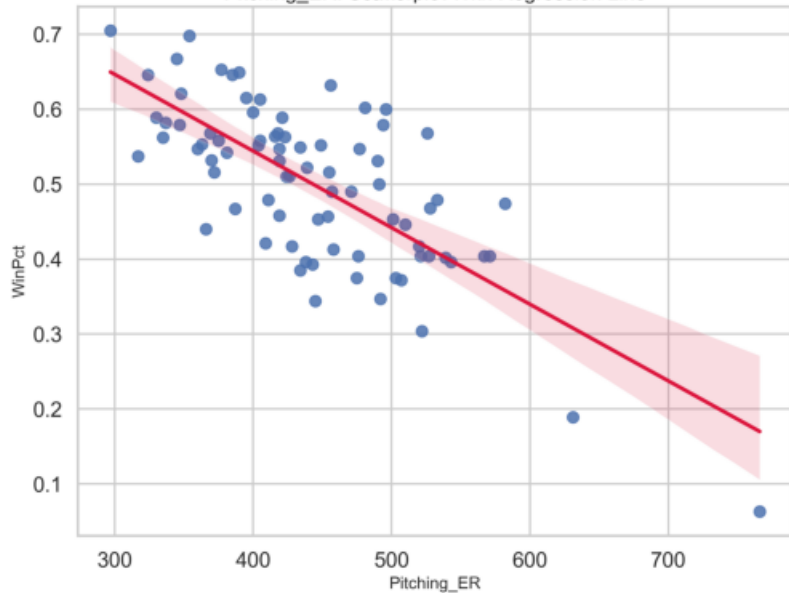
Interpretation

Pitching_ER has a statistically significant negative relationship with WinPct and explains approximately 53.9% of the variation in team winning percentage. Pitching_ER residuals do not exhibit significant heteroskedasticity under either the Breusch-Pagan or White tests. Residual normality is not rejected at the 5% level.

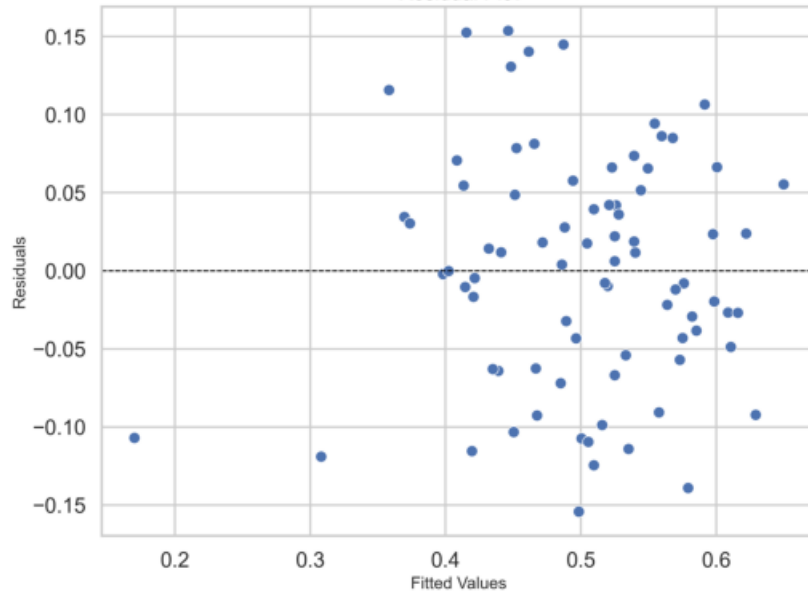
Warnings

- 4 observations exceed the leverage threshold of 0.0500.
- 4 observations exceed the Cook's distance threshold of 0.0500.

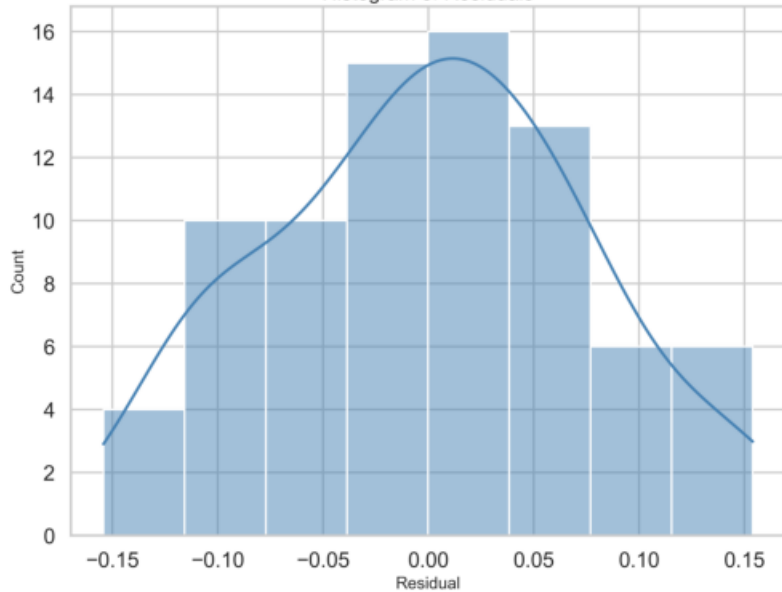
Pitching_ER: Scatterplot With Regression Line



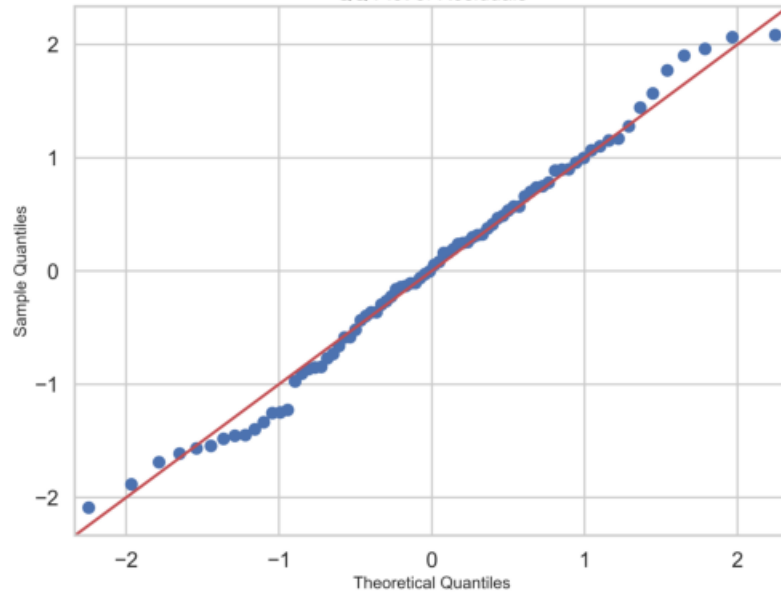
Residual Plot



Histogram of Residuals



QQ Plot of Residuals



Pitching_SV Regression Summary

Formula: Q("Pitching_W-L%") ~ Q("Pitching_SV")

Coefficient Table

term	coef	std_err	t_stat	p_value	ci_lower	ci_upper
Intercept	0.2429	0.0324	7.4905	0.0	0.1783	0.3075
Q("Pitching_SV")	0.0127	0.0015	8.2708	0.0	0.0096	0.0157

Model Fit

	value
R-squared	0.4672
Adjusted R-squared	0.4604
F-statistic	68.4054
F-statistic p-value	0.0000
Observations	80.0000

Heteroskedasticity Tests

Test	LM Statistic	LM p-value	F Statistic	F p-value
Breusch-Pagan	1.9964	0.1577	1.9963	0.1617
White	5.8877	0.0527	3.0586	0.0527

Normality Test

JB statistic = 0.6799, JB p-value = 0.7118, Skewness = -0.0140, Kurtosis = 2.5492

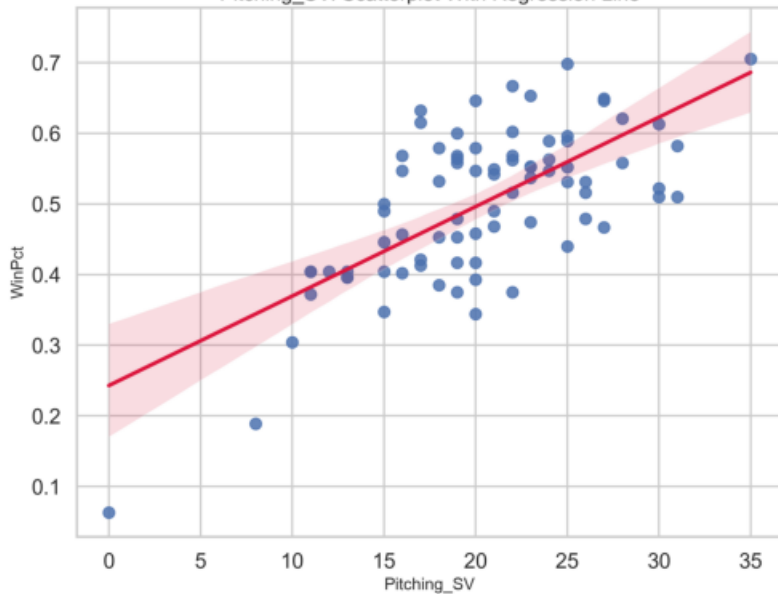
Interpretation

Pitching_SV has a statistically significant positive relationship with WinPct and explains approximately 46.7% of the variation in team winning percentage. Pitching_SV residuals do not exhibit significant heteroskedasticity under either the Breusch-Pagan or White tests. Residual normality is not rejected at the 5% level.

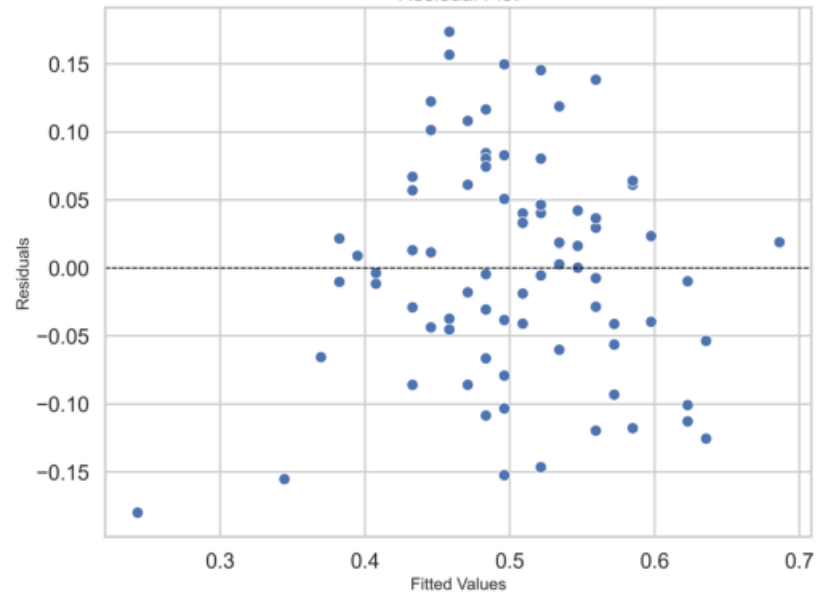
Warnings

- 6 observations exceed the leverage threshold of 0.0500.
- 4 observations exceed the Cook's distance threshold of 0.0500.

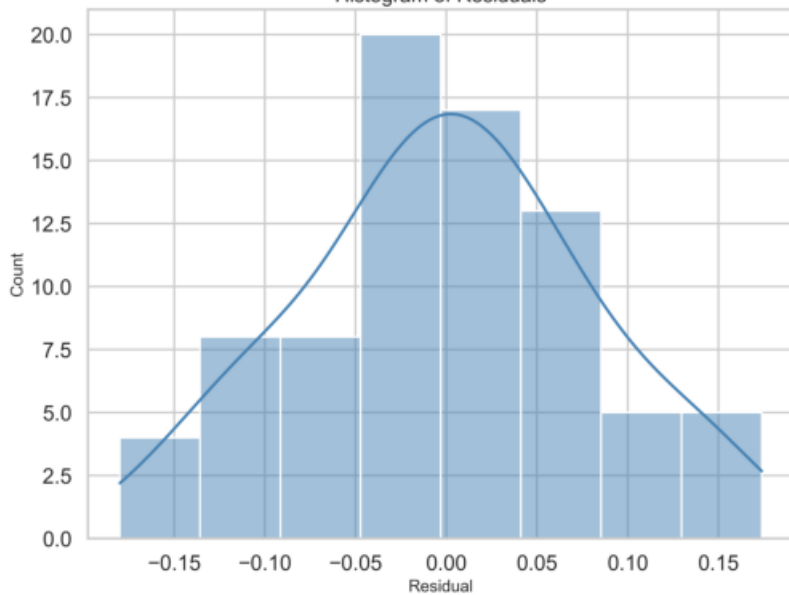
Pitching_SV: Scatterplot With Regression Line



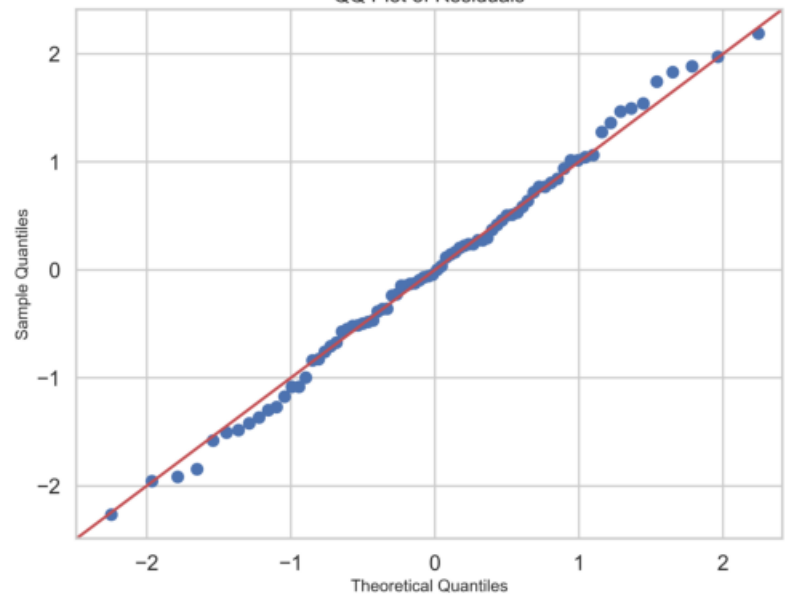
Residual Plot



Histogram of Residuals



QQ Plot of Residuals



FIP Regression Summary

Formula: Q("Pitching_W-L%") ~ Q("FIP")

Coefficient Table						
	coef	std_err	t_stat	p_value	ci_lower	ci_upper
term						
Intercept	1.1066	0.0821	13.4795	0.0	0.9432	1.2700
Q("FIP")	-0.1244	0.0167	-7.4309	0.0	-0.1578	-0.0911

Model Fit	
	value
R-squared	0.4145
Adjusted R-squared	0.4070
F-statistic	55.2188
F-statistic p-value	0.0000
Observations	80.0000

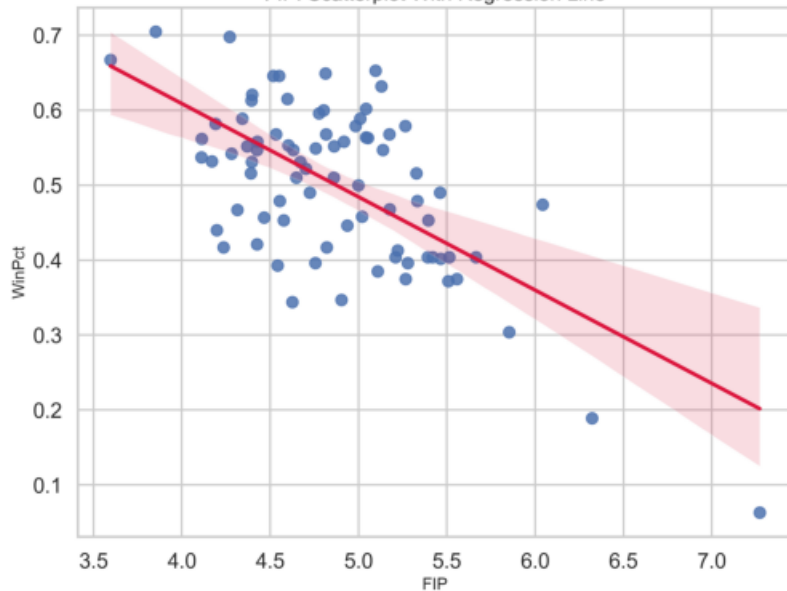
Heteroskedasticity Tests					
	Test	LM Statistic	LM p-value	F Statistic	F p-value
Breusch-Pagan		0.8068	0.3691	0.7947	0.3754
White		1.8667	0.3932	0.9198	0.4029

Normality Test
JB statistic = 1.3839, JB p-value = 0.5006, Skewness = -0.0868, Kurtosis = 2.3795

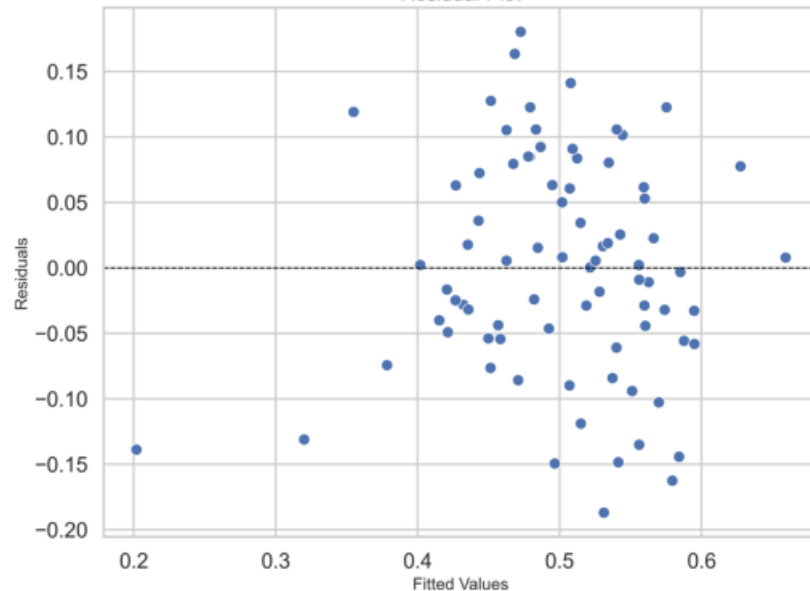
Interpretation
FIP has a statistically significant negative relationship with WinPct and explains approximately 41.4% of the variation in team winning percentage. FIP residuals do not exhibit significant heteroskedasticity under either the Breusch-Pagan or White tests. Residual normality is not rejected at the 5% level.

Warnings
6 observations exceed the leverage threshold of 0.0500.
4 observations exceed the Cook's distance threshold of 0.0500.

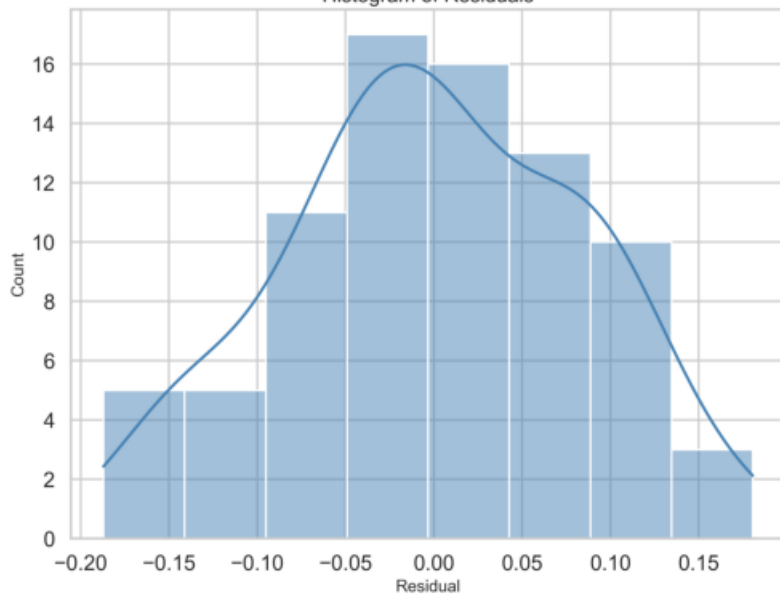
FIP: Scatterplot With Regression Line



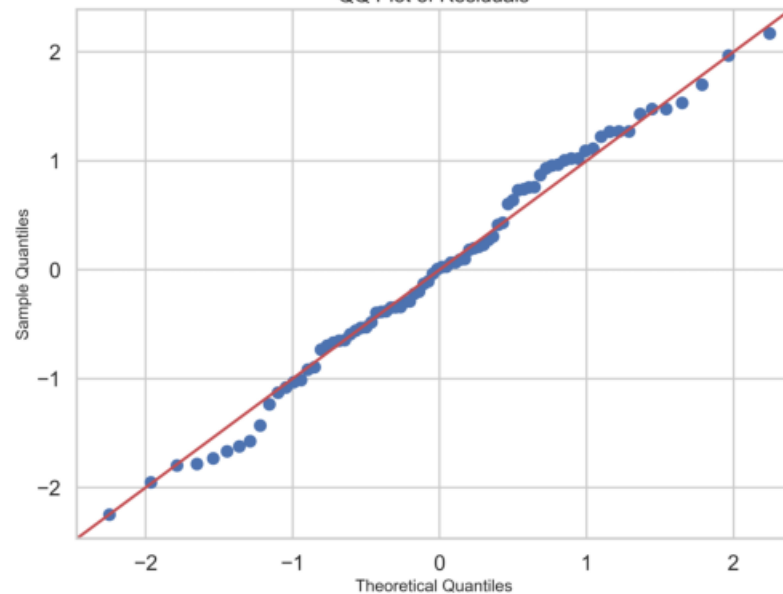
Residual Plot



Histogram of Residuals



QQ Plot of Residuals



Batting_H Regression Summary

Formula: Q("Pitching_W-L%") ~ Q("Batting_H")

Coefficient Table

	coef	std_err	t_stat	p_value	ci_lower	ci_upper
term						
Intercept	-0.3367	0.1366	-2.4658	0.0159	-0.6086	-0.0649
Q("Batting_H")	0.0010	0.0002	6.1483	0.0000	0.0007	0.0013

Model Fit

	value
R-squared	0.3264
Adjusted R-squared	0.3178
F-statistic	37.8014
F-statistic p-value	0.0000
Observations	80.0000

Heteroskedasticity Tests

Test	LM Statistic	LM p-value	F Statistic	F p-value
Breusch-Pagan	3.7576	0.0526	3.8442	0.0535
White	5.5910	0.0611	2.8929	0.0615

Normality Test

JB statistic = 7.1780, JB p-value = 0.0276, Skewness = -0.4754, Kurtosis = 4.1178

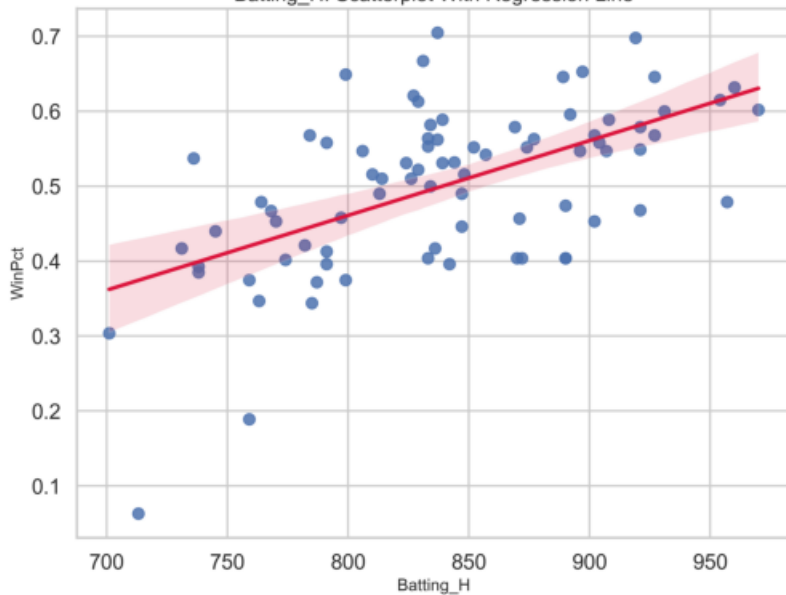
Interpretation

Batting_H has a statistically significant positive relationship with WinPct and explains approximately 32.6% of the variation in team winning percentage. Batting_H residuals do not exhibit significant heteroskedasticity under either the Breusch-Pagan or White tests. Residual normality is rejected at the 5% level.

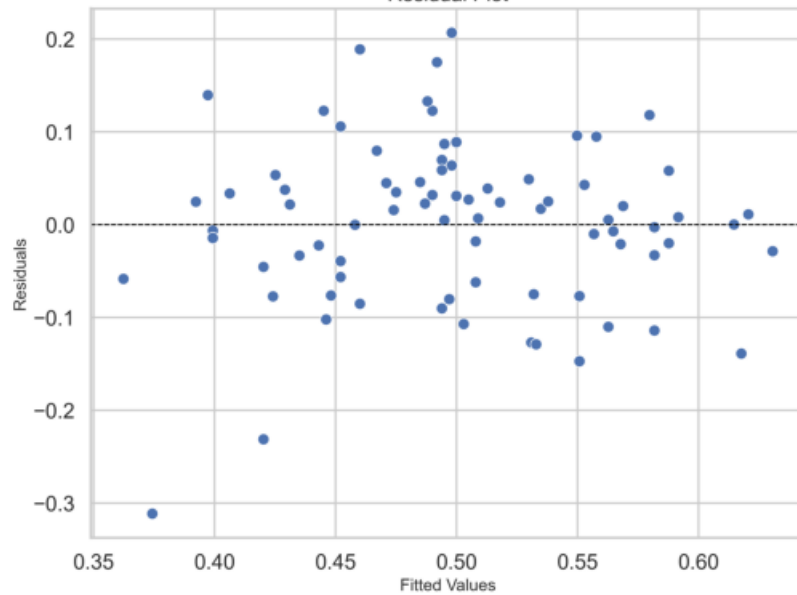
Warnings

- 7 observations exceed the leverage threshold of 0.0500.
- 4 observations exceed the Cook's distance threshold of 0.0500.

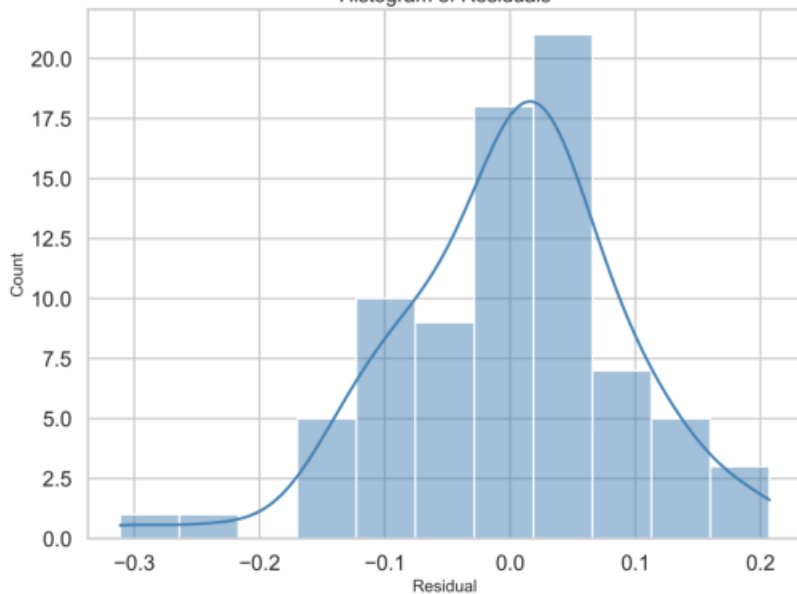
Batting_H: Scatterplot With Regression Line



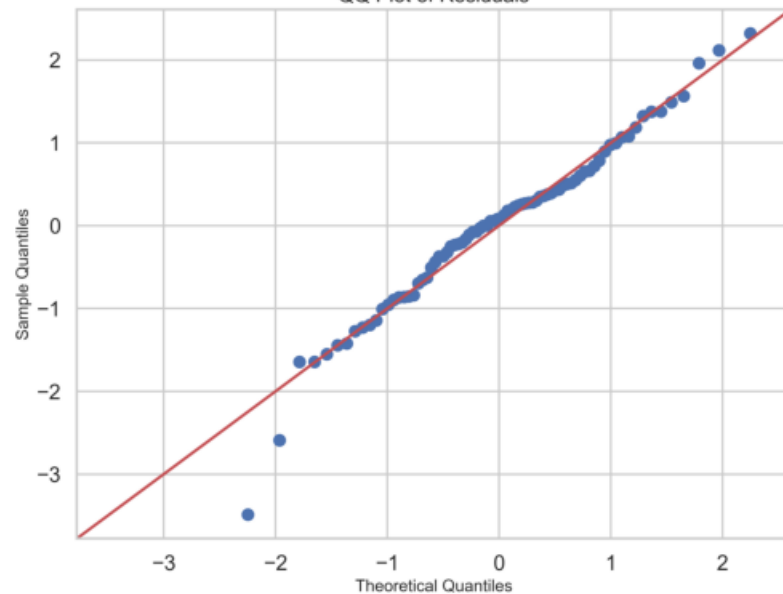
Residual Plot



Histogram of Residuals



QQ Plot of Residuals



SLG Regression Summary

Formula: Q("Pitching_W-L%") ~ Q("Batting_SLG")

Coefficient Table

term	coef	std_err	t_stat	p_value	ci_lower	ci_upper
Intercept	-0.1182	0.1061	-1.1145	0.2685	-0.3295	0.0930
Q("Batting_SLG")	1.5217	0.2597	5.8600	0.0000	1.0047	2.0387

Model Fit

	value
R-squared	0.3057
Adjusted R-squared	0.2968
F-statistic	34.3394
F-statistic p-value	0.0000
Observations	80.0000

Heteroskedasticity Tests

Test	LM Statistic	LM p-value	F Statistic	F p-value
Breusch-Pagan	2.2264	0.1357	2.2329	0.1391
White	4.8844	0.0870	2.5035	0.0884

Normality Test

JB statistic = 6.6403, JB p-value = 0.0361, Skewness = -0.6469, Kurtosis = 3.5641

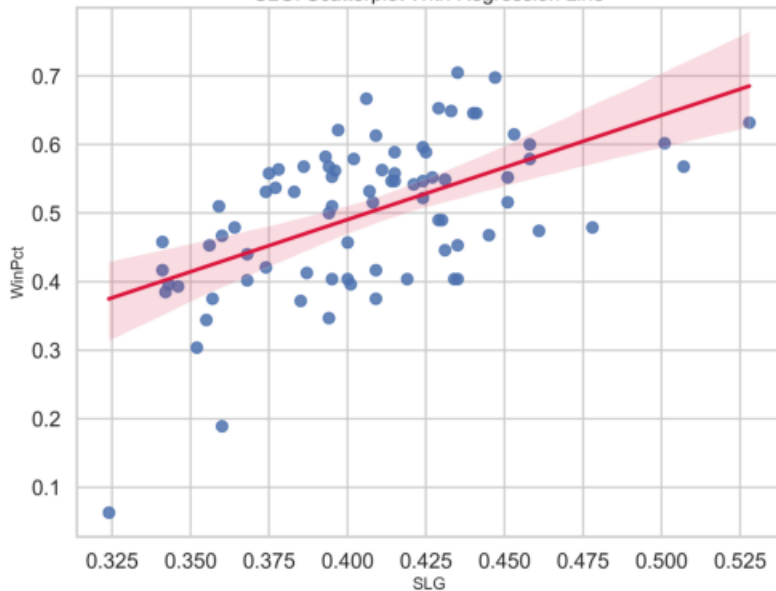
Interpretation

SLG has a statistically significant positive relationship with WinPct and explains approximately 30.6% of the variation in team winning percentage. SLG residuals do not exhibit significant heteroskedasticity under either the Breusch-Pagan or White tests. Residual normality is rejected at the 5% level.

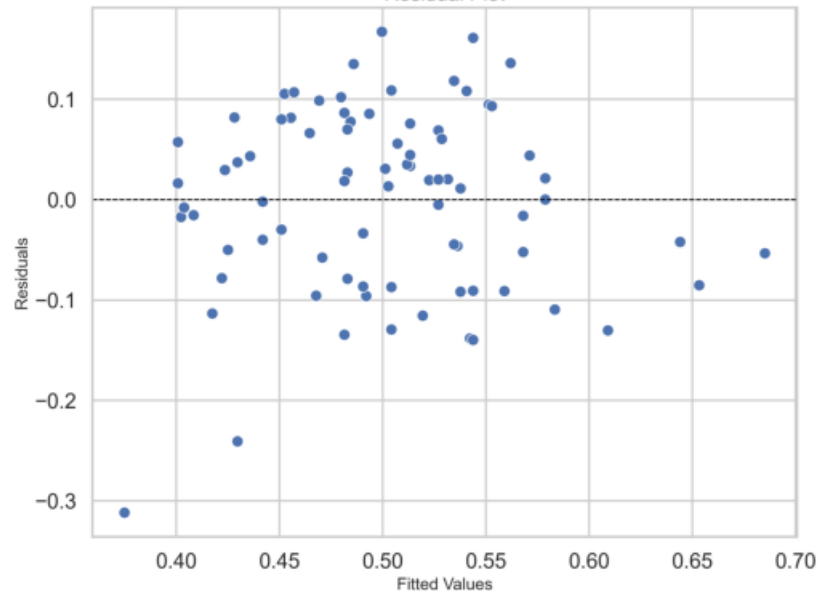
Warnings

- 5 observations exceed the leverage threshold of 0.0500.
- 3 observations exceed the Cook's distance threshold of 0.0500.

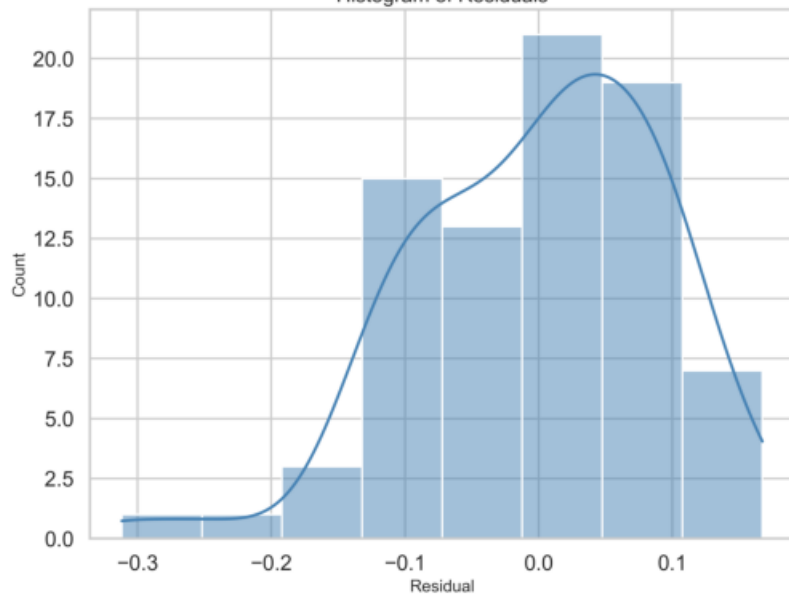
SLG: Scatterplot With Regression Line



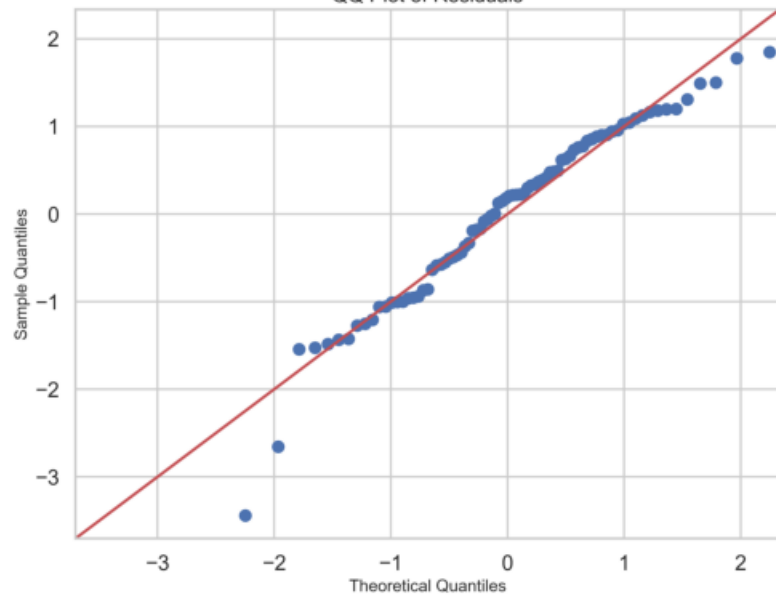
Residual Plot



Histogram of Residuals



QQ Plot of Residuals



Batting_TB Regression Summary

Formula: Q("Pitching_W-L%") ~ Q("Batting_TB")

Coefficient Table

	coef	std_err	t_stat	p_value	ci_lower	ci_upper
term						
Intercept	-0.0365	0.0951	-0.3835	0.7024	-0.2258	0.1528
Q("Batting_TB")	0.0004	0.0001	5.6821	0.0000	0.0003	0.0006

Model Fit

	value
R-squared	0.2927
Adjusted R-squared	0.2837
F-statistic	32.2857
F-statistic p-value	0.0000
Observations	80.0000

Heteroskedasticity Tests

Test	LM Statistic	LM p-value	F Statistic	F p-value
Breusch-Pagan	2.1137	0.1460	2.1168	0.1497
White	3.5890	0.1662	1.8083	0.1708

Normality Test

JB statistic = 8.5564, JB p-value = 0.0139, Skewness = -0.6667, Kurtosis = 3.8883

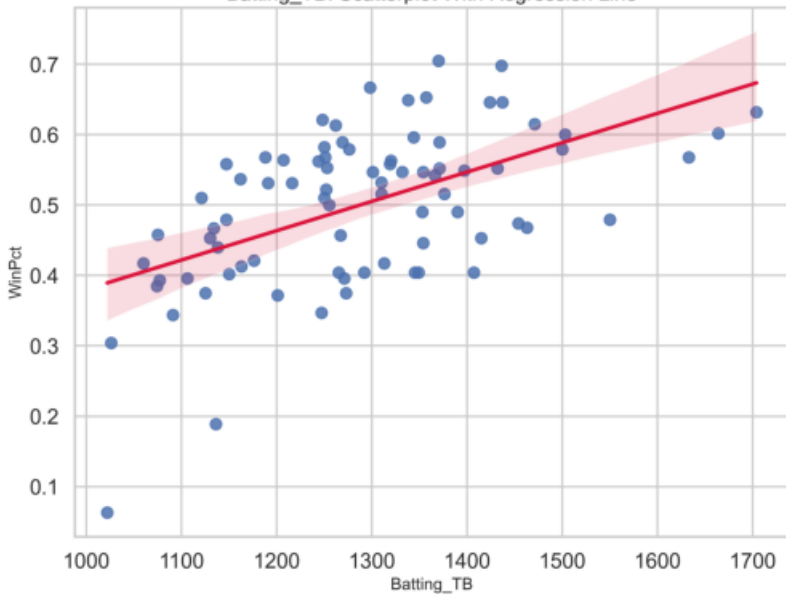
Interpretation

Batting_TB has a statistically significant positive relationship with WinPct and explains approximately 29.3% of the variation in team winning percentage. Batting_TB residuals do not exhibit significant heteroskedasticity under either the Breusch-Pagan or White tests. Residual normality is rejected at the 5% level.

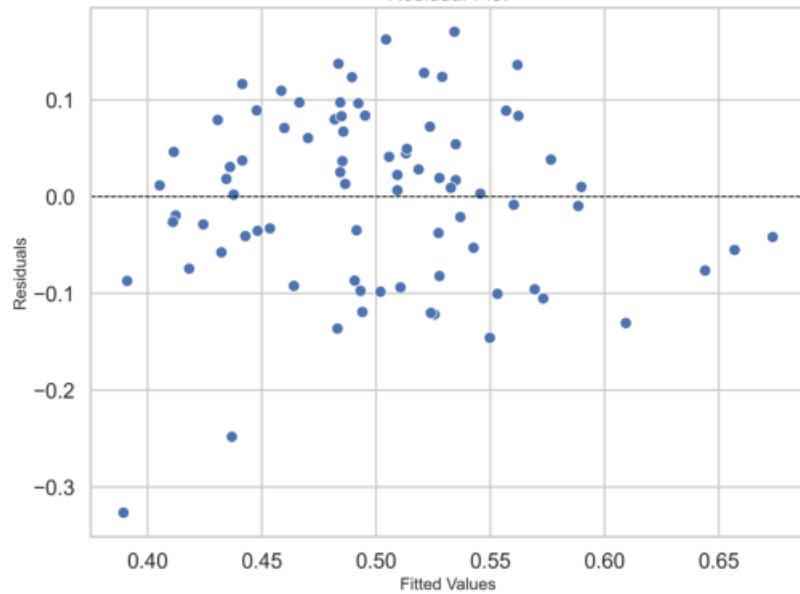
Warnings

- 6 observations exceed the leverage threshold of 0.0500.
- 3 observations exceed the Cook's distance threshold of 0.0500.

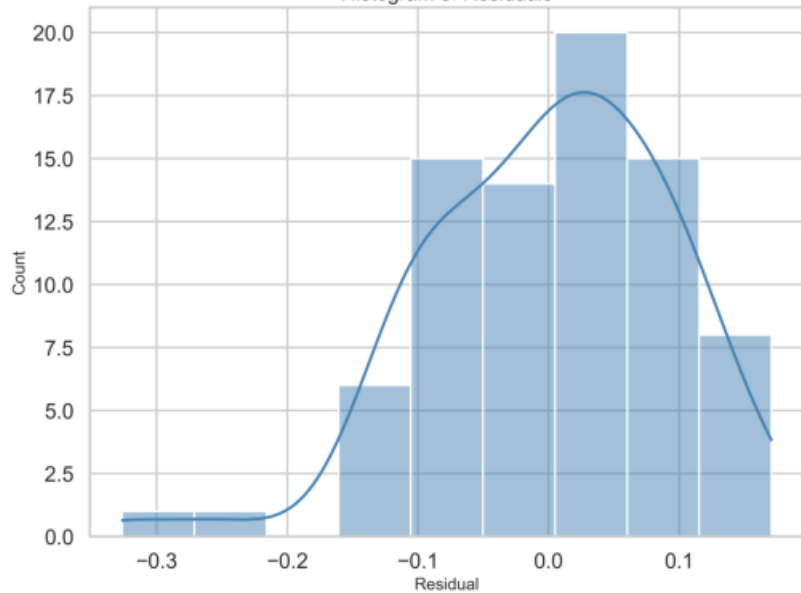
Batting_TB: Scatterplot With Regression Line



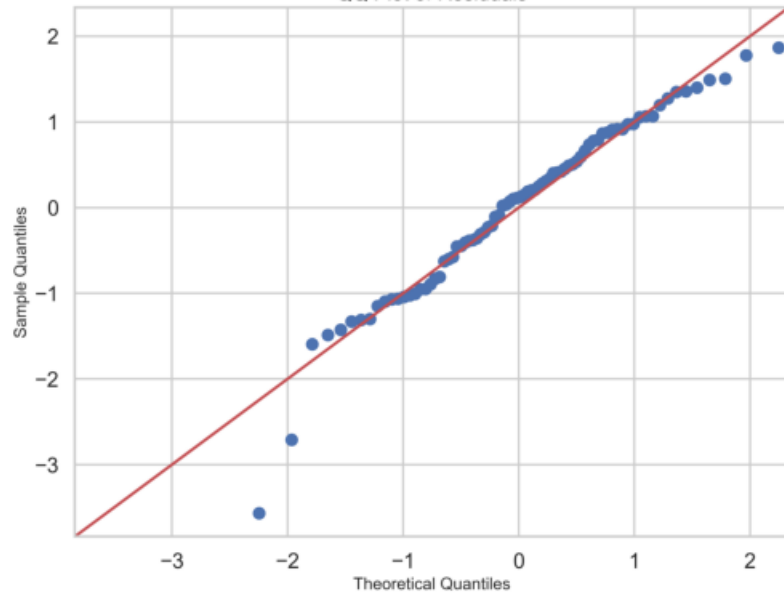
Residual Plot



Histogram of Residuals



QQ Plot of Residuals



Batting_SF Regression Summary

Formula: Q("Pitching_W-L%") ~ Q("Batting_SF")

Coefficient Table

term	coef	std_err	t_stat	p_value	ci_lower	ci_upper
Intercept	0.2862	0.0458	6.2524	0.0	0.1951	0.3773
Q("Batting_SF")	0.0063	0.0013	4.8196	0.0	0.0037	0.0089

Model Fit

	value
R-squared	0.2295
Adjusted R-squared	0.2196
F-statistic	23.2289
F-statistic p-value	0.0000
Observations	80.0000

Heteroskedasticity Tests

Test	LM Statistic	LM p-value	F Statistic	F p-value
Breusch-Pagan	2.8760	0.0899	2.9086	0.0921
White	3.3685	0.1856	1.6923	0.1909

Normality Test

JB statistic = 15.9583, JB p-value = 0.0003, Skewness = -0.6819, Kurtosis = 4.7110

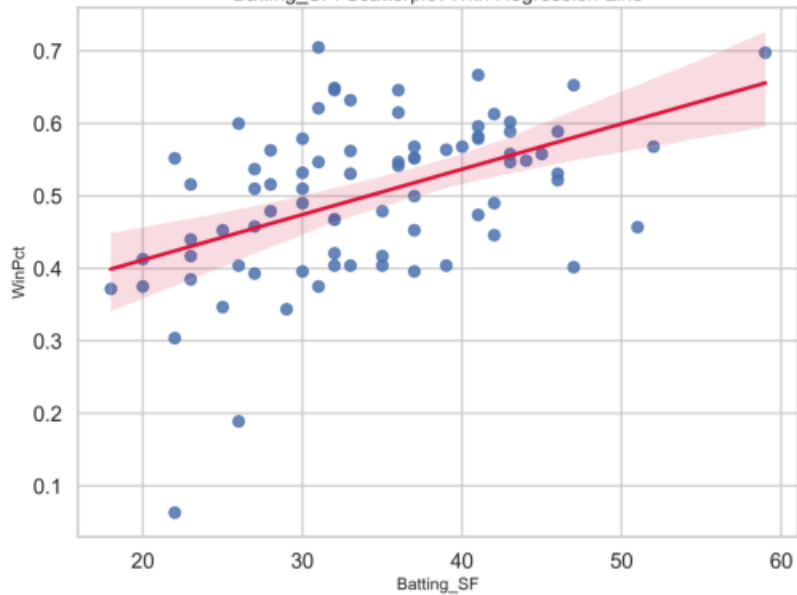
Interpretation

Batting_SF has a statistically significant positive relationship with WinPct and explains approximately 22.9% of the variation in team winning percentage. Batting_SF residuals do not exhibit significant heteroskedasticity under either the Breusch-Pagan or White tests. Residual normality is rejected at the 5% level.

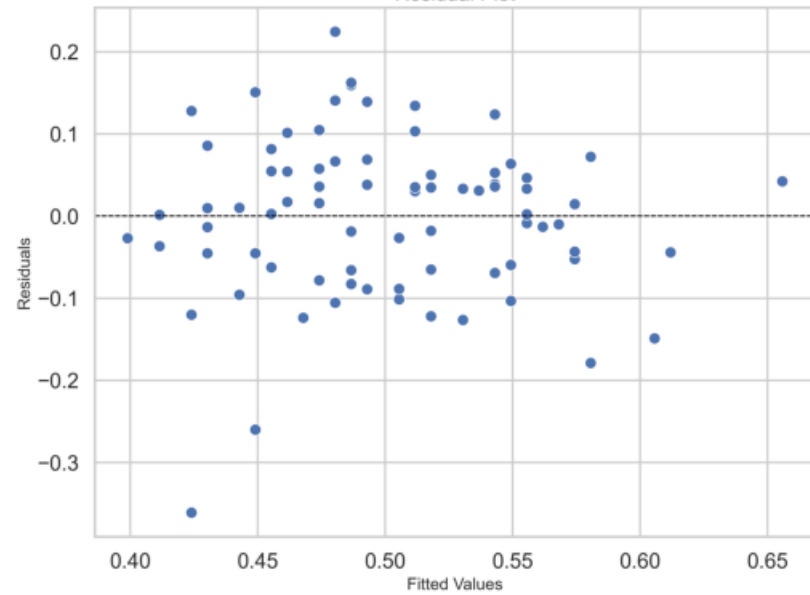
Warnings

- 4 observations exceed the leverage threshold of 0.0500.
- 4 observations exceed the Cook's distance threshold of 0.0500.

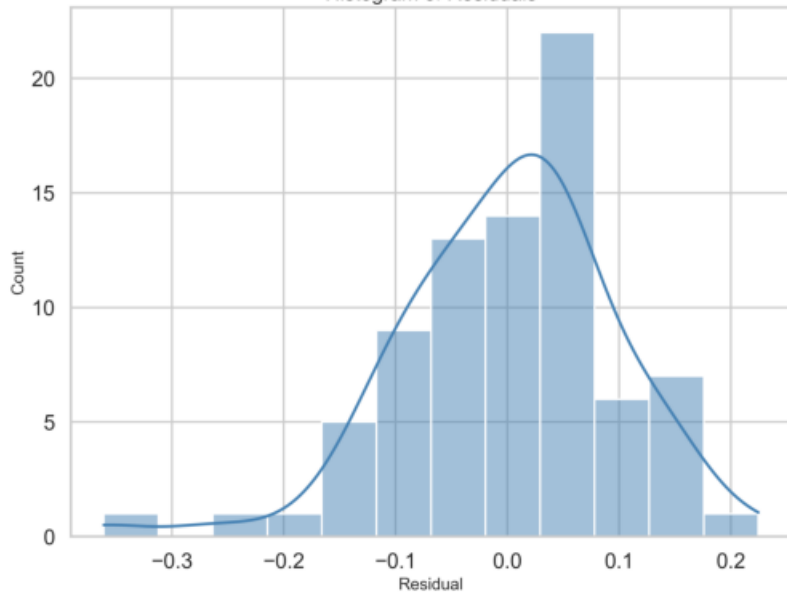
Batting_SF: Scatterplot With Regression Line



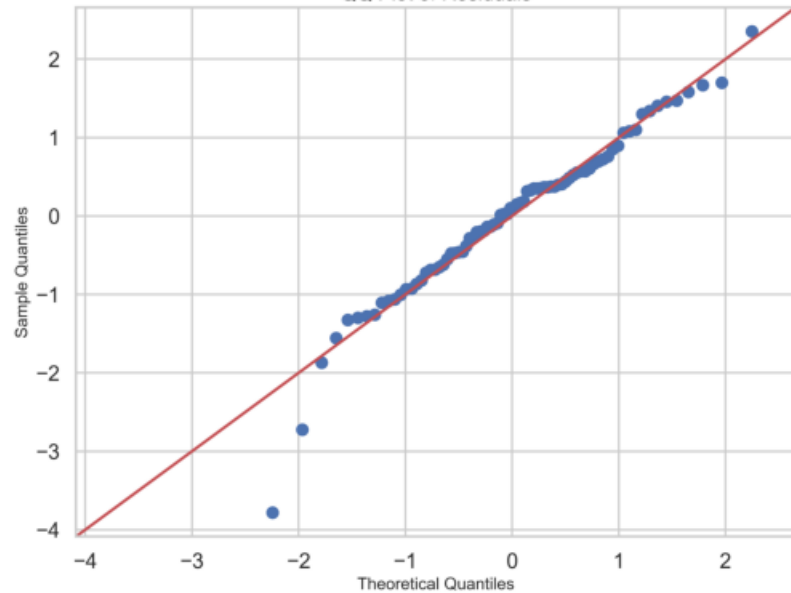
Residual Plot



Histogram of Residuals



QQ Plot of Residuals



HR Regression Summary

Formula: Q("Pitching_W-L%") ~ Q("Batting_HR")

Coefficient Table

term	coef	std_err	t_stat	p_value	ci_lower	ci_upper
Intercept	0.3574	0.0392	9.129	0.0000	0.2795	0.4354
Q("Batting_HR")	0.0016	0.0004	3.819	0.0003	0.0008	0.0025

Model Fit

	value
R-squared	0.1575
Adjusted R-squared	0.1467
F-statistic	14.5849
F-statistic p-value	0.0003
Observations	80.0000

Heteroskedasticity Tests

Test	LM Statistic	LM p-value	F Statistic	F p-value
Breusch-Pagan	2.1785	0.1399	2.1835	0.1435
White	2.4024	0.3008	1.1920	0.3092

Normality Test

JB statistic = 15.4325, JB p-value = 0.0004, Skewness = -0.8224, Kurtosis = 4.3871

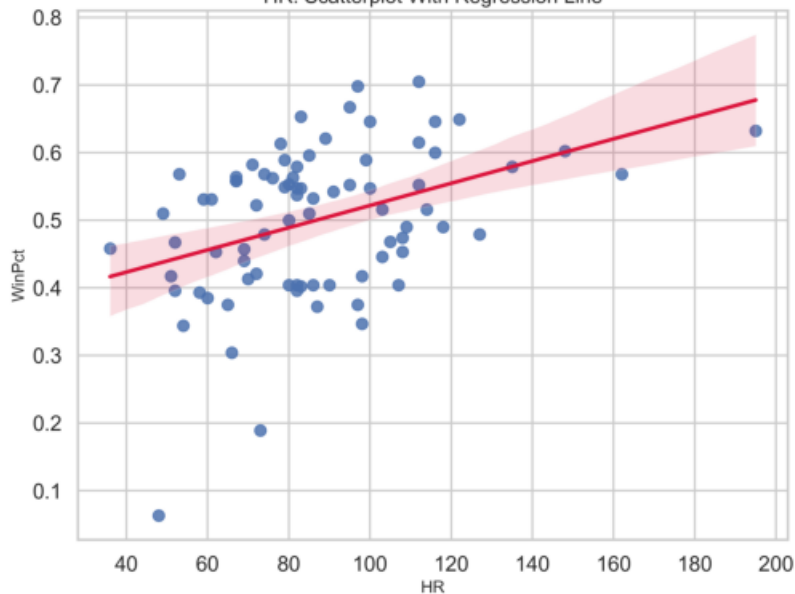
Interpretation

HR has a statistically significant positive relationship with WinPct and explains approximately 15.8% of the variation in team winning percentage. HR residuals do not exhibit significant heteroskedasticity under either the Breusch-Pagan or White tests. Residual normality is rejected at the 5% level.

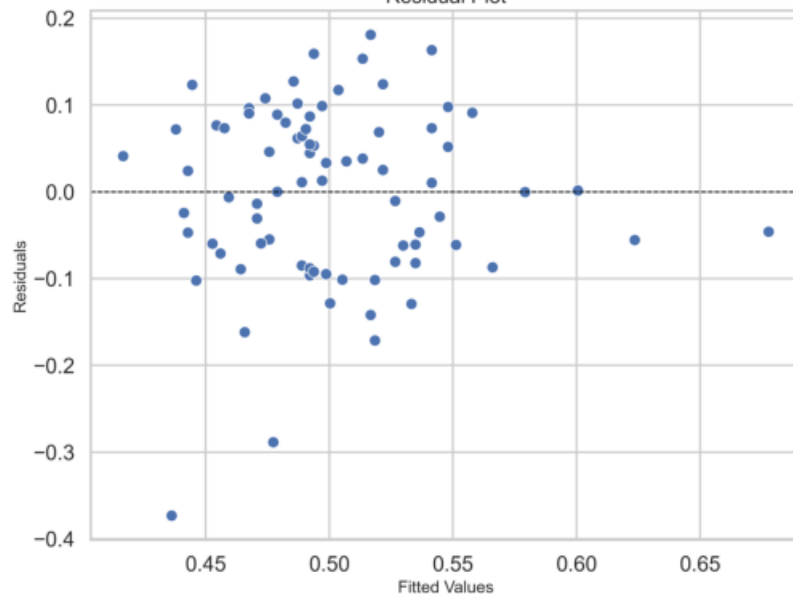
Warnings

- 5 observations exceed the leverage threshold of 0.0500.
- 2 observations exceed the Cook's distance threshold of 0.0500.

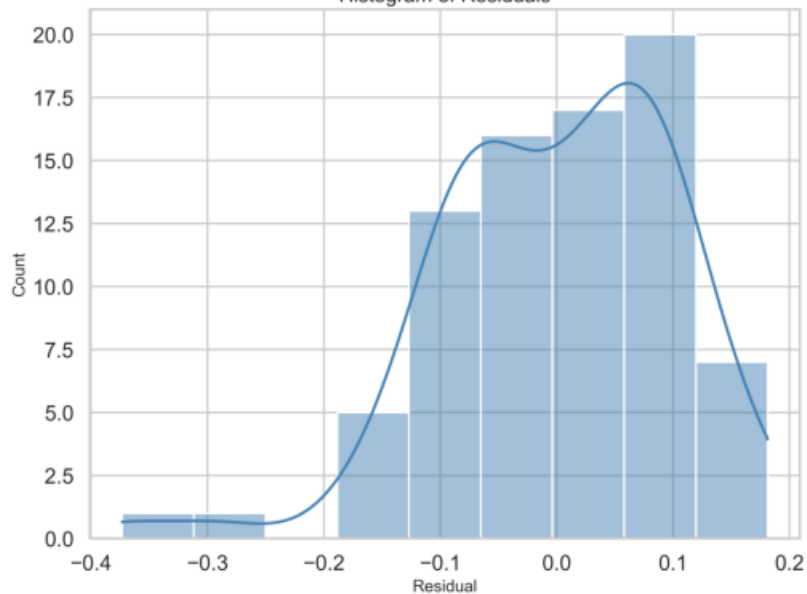
HR: Scatterplot With Regression Line



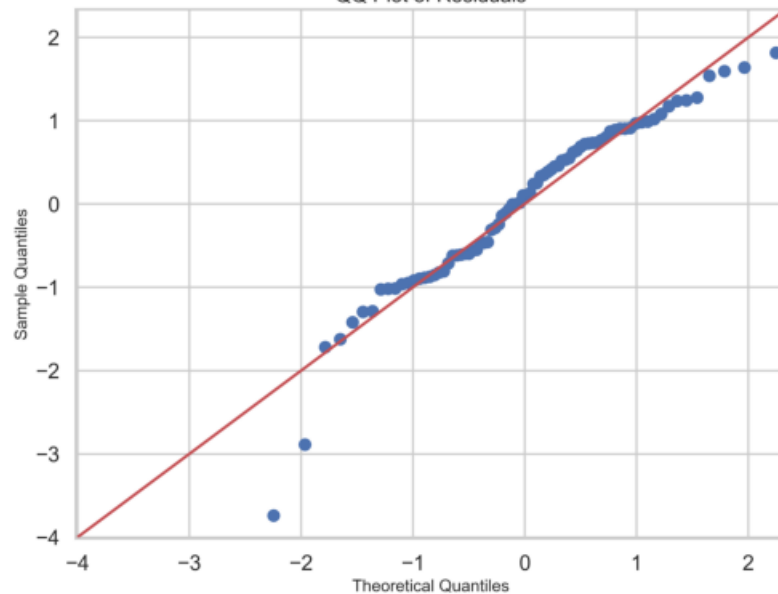
Residual Plot



Histogram of Residuals



QQ Plot of Residuals



Batting_BatAge Regression Summary

Formula: Q("Pitching_W-L%") ~ Q("Batting_BatAge")

Coefficient Table

	coef	std_err	t_stat	p_value	ci_lower	ci_upper
term						
Intercept	-0.7716	0.3909	-1.9739	0.0519	-1.5499	0.0066
Q("Batting_BatAge")	0.0496	0.0152	3.2559	0.0017	0.0193	0.0800

Model Fit

	value
R-squared	0.1196
Adjusted R-squared	0.1084
F-statistic	10.6009
F-statistic p-value	0.0017
Observations	80.0000

Heteroskedasticity Tests

Test	LM Statistic	LM p-value	F Statistic	F p-value
Breusch-Pagan	0.7728	0.3794	0.7608	0.3858
White	0.7794	0.6772	0.3788	0.6860

Normality Test

JB statistic = 21.1833, JB p-value = 0.0000, Skewness = -0.8631, Kurtosis = 4.8373

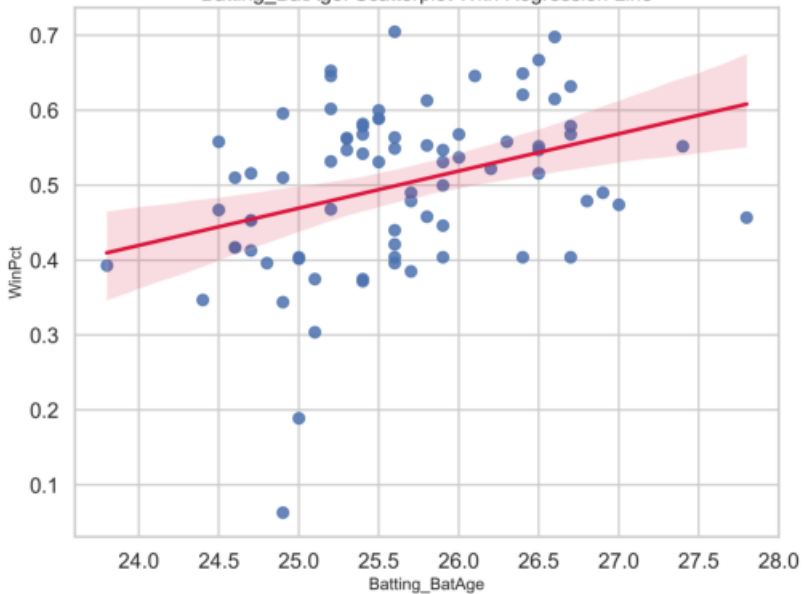
Interpretation

Batting_BatAge has a statistically significant positive relationship with WinPct and explains approximately 12.0% of the variation in team winning percentage. Batting_BatAge residuals do not exhibit significant heteroskedasticity under either the Breusch-Pagan or White tests. Residual normality is rejected at the 5% level.

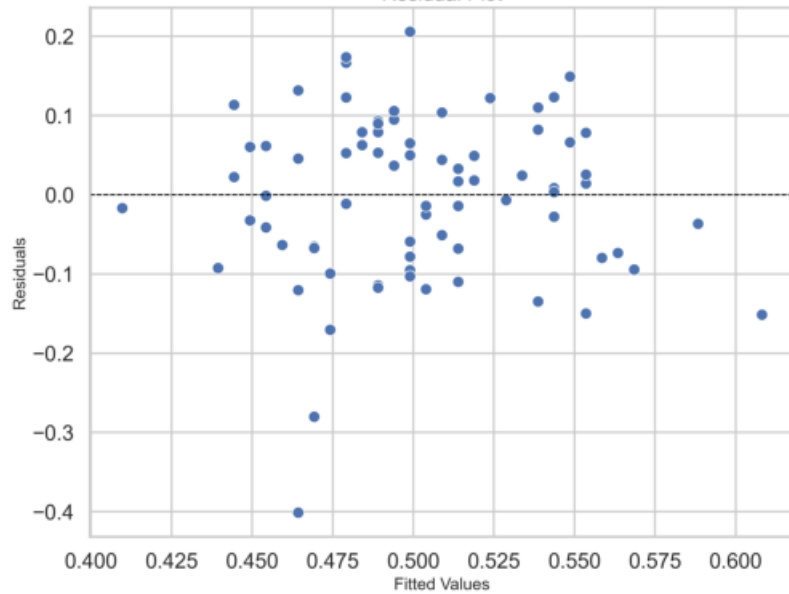
Warnings

- 4 observations exceed the leverage threshold of 0.0500.
- 3 observations exceed the Cook's distance threshold of 0.0500.

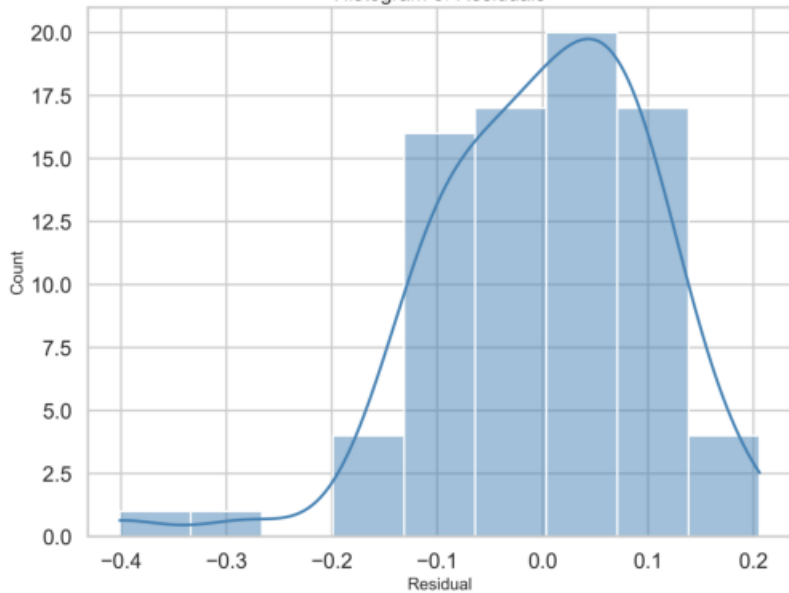
Batting_BatAge: Scatterplot With Regression Line



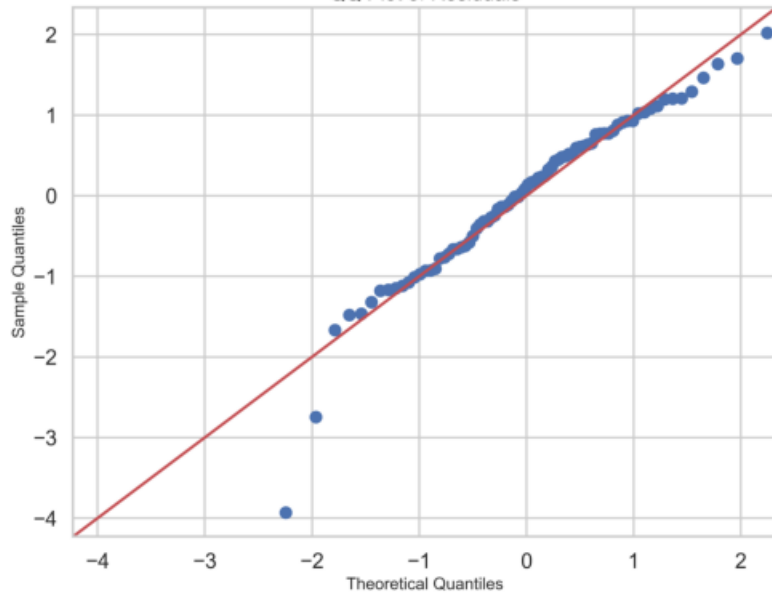
Residual Plot



Histogram of Residuals



QQ Plot of Residuals



S09 Regression Summary

Formula: Q("Pitching_W-L%") ~ Q("Pitching_S09")

Coefficient Table

term	coef	std_err	t_stat	p_value	ci_lower	ci_upper
Intercept	0.1647	0.1247	1.3215	0.1902	-0.0834	0.4129
Q("Pitching_S09")	0.0413	0.0153	2.7060	0.0084	0.0109	0.0717

Model Fit

	value
R-squared	0.0858
Adjusted R-squared	0.0741
F-statistic	7.3226
F-statistic p-value	0.0084
Observations	80.0000

Heteroskedasticity Tests

Test	LM Statistic	LM p-value	F Statistic	F p-value
Breusch-Pagan	3.5786	0.0585	3.6525	0.0597
White	4.6391	0.0983	2.3700	0.1003

Normality Test

JB statistic = 12.9149, JB p-value = 0.0016, Skewness = -0.7458, Kurtosis = 4.2844

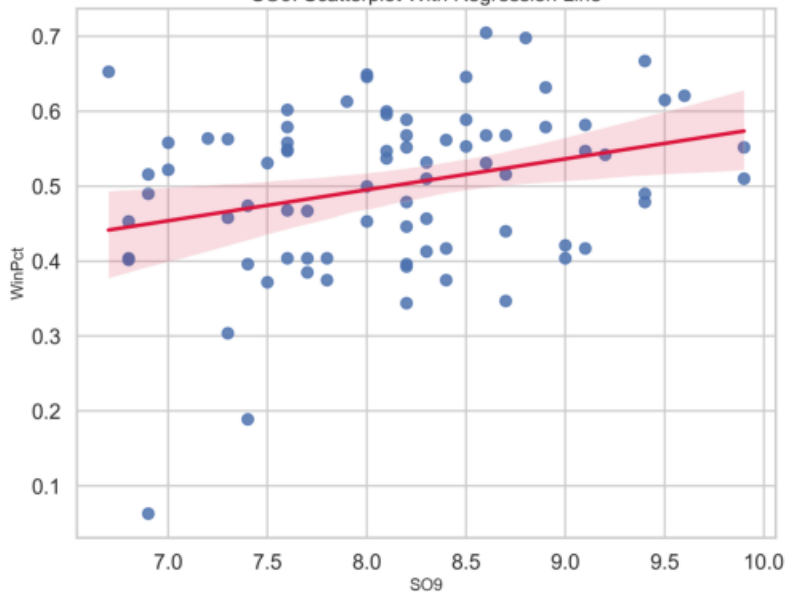
Interpretation

S09 has a statistically significant positive relationship with WinPct and explains approximately 8.6% of the variation in team winning percentage. S09 residuals do not exhibit significant heteroskedasticity under either the Breusch-Pagan or White tests. Residual normality is rejected at the 5% level.

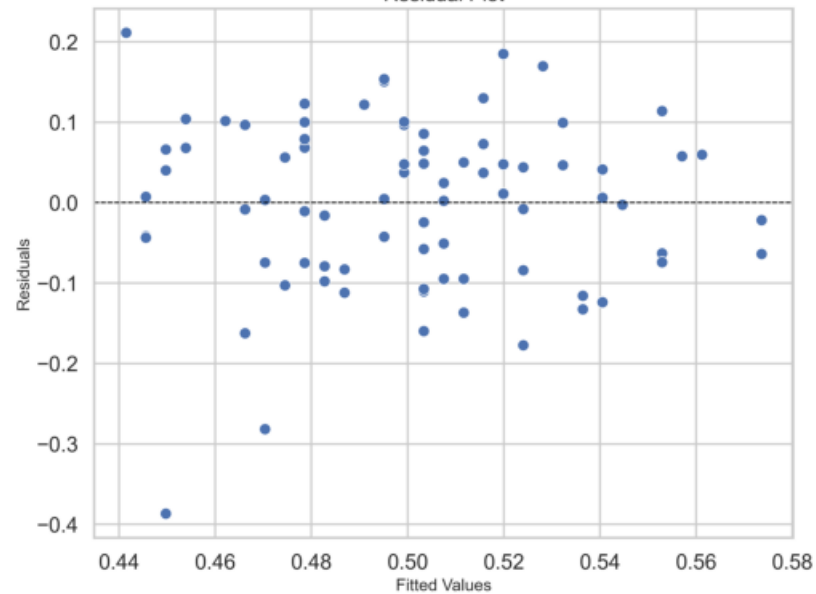
Warnings

- 5 observations exceed the leverage threshold of 0.0500.
- 3 observations exceed the Cook's distance threshold of 0.0500.

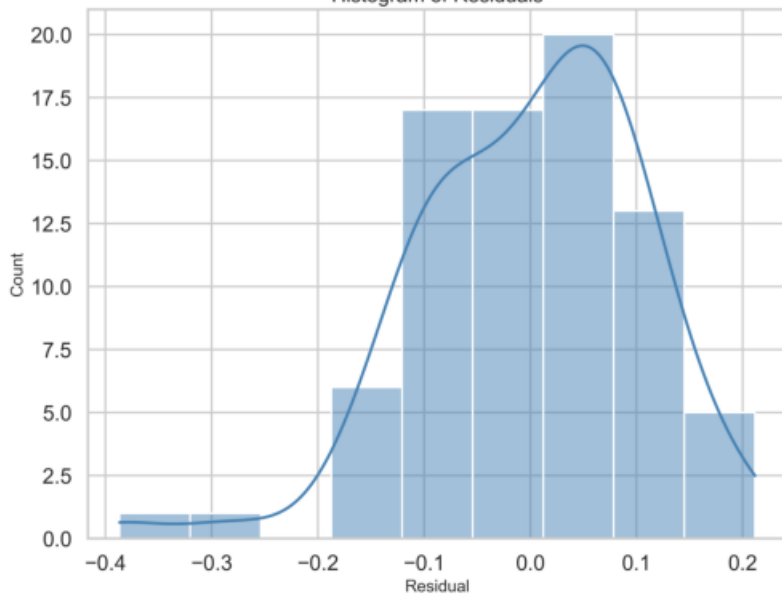
SO9: Scatterplot With Regression Line



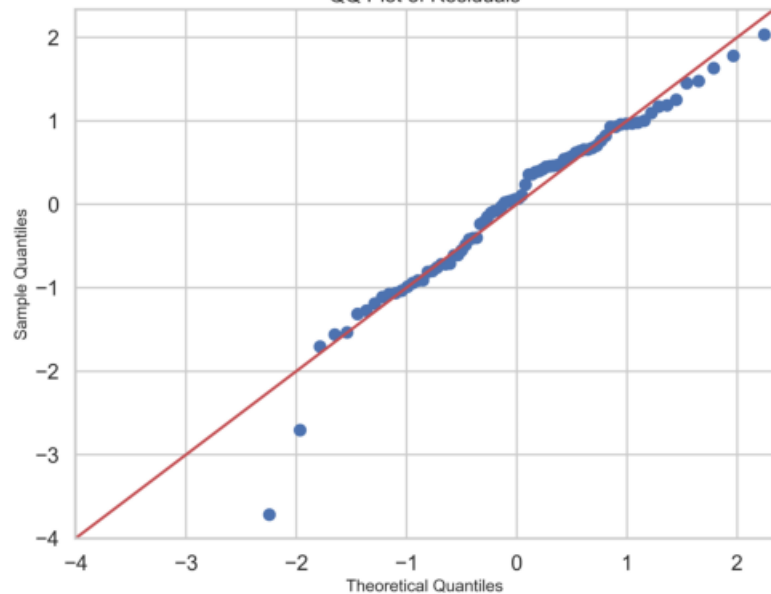
Residual Plot



Histogram of Residuals



QQ Plot of Residuals



Batting_AB Regression Summary

Formula: Q("Pitching_W-L%") ~ Q("Batting_AB")

Coefficient Table

	coef	std_err	t_stat	p_value	ci_lower	ci_upper
term						
Intercept	-0.6922	0.5231	-1.3232	0.1896	-1.7336	0.3492
Q("Batting_AB")	0.0004	0.0002	2.2808	0.0253	0.0000	0.0007

Model Fit

	value
R-squared	0.0625
Adjusted R-squared	0.0505
F-statistic	5.2021
F-statistic p-value	0.0253
Observations	80.0000

Heteroskedasticity Tests

Test	LM Statistic	LM p-value	F Statistic	F p-value
Breusch-Pagan	0.5623	0.4534	0.5521	0.4597
White	1.5917	0.4512	0.7816	0.4613

Normality Test

JB statistic = 32.6724, JB p-value = 0.0000, Skewness = -0.9356, Kurtosis = 5.5100

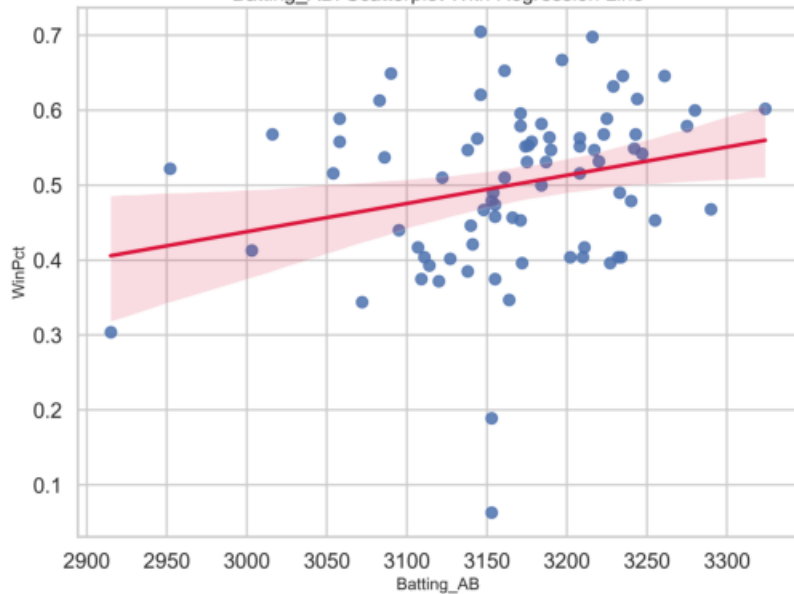
Interpretation

Batting_AB has a statistically significant positive relationship with WinPct and explains approximately 6.3% of the variation in team winning percentage. Batting_AB residuals do not exhibit significant heteroskedasticity under either the Breusch-Pagan or White tests. Residual normality is rejected at the 5% level.

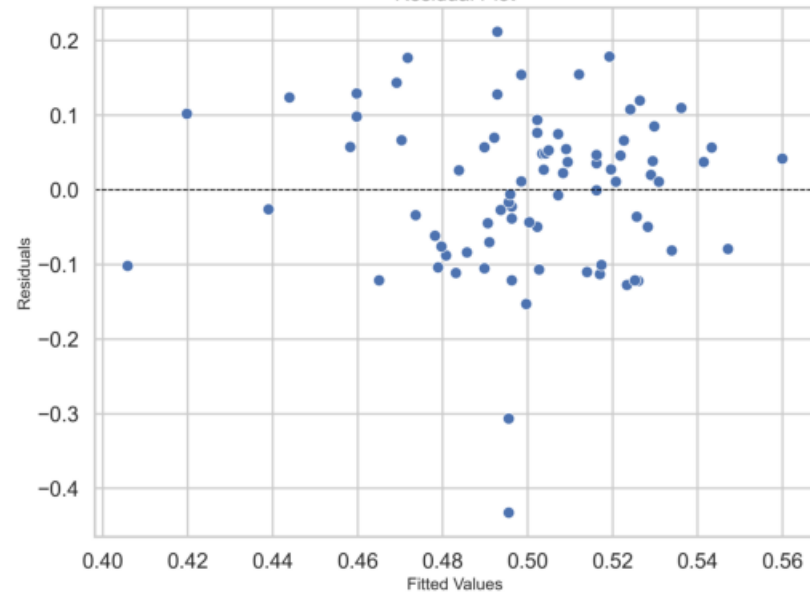
Warnings

- 5 observations exceed the leverage threshold of 0.0500.
- 5 observations exceed the Cook's distance threshold of 0.0500.

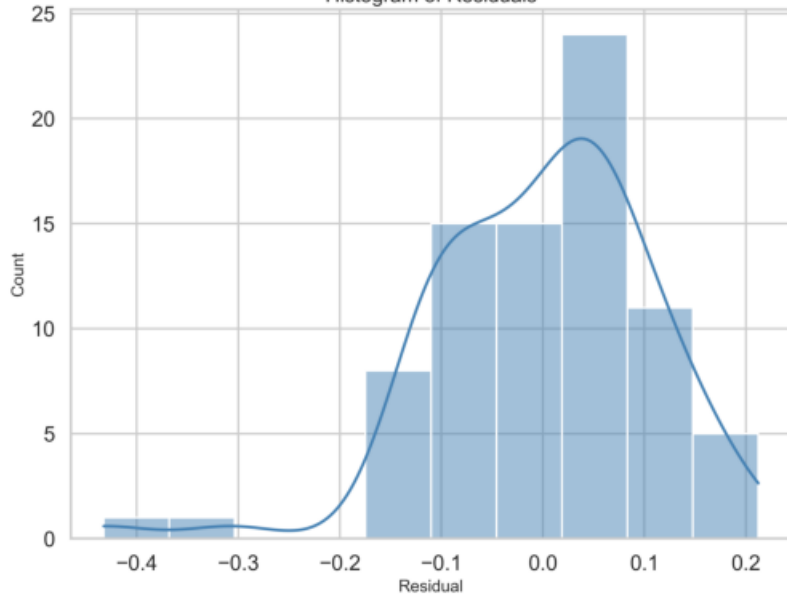
Batting_AB: Scatterplot With Regression Line



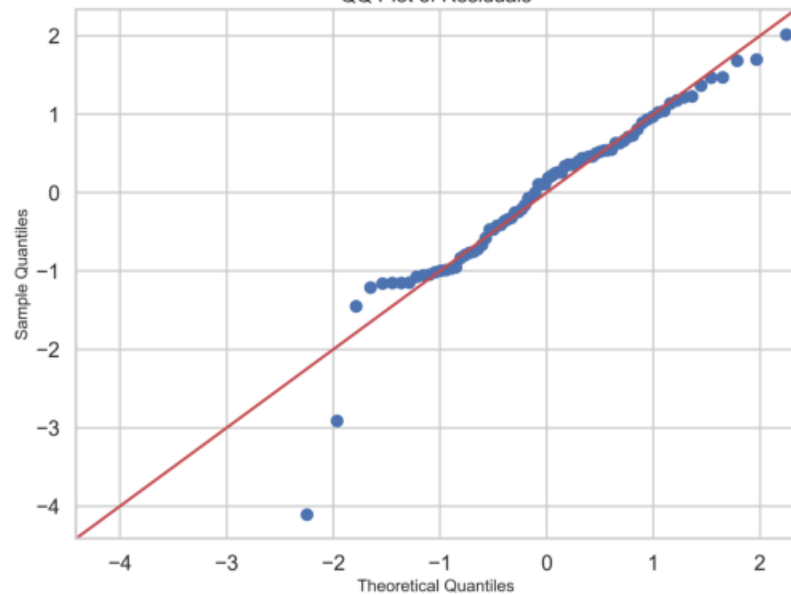
Residual Plot



Histogram of Residuals



QQ Plot of Residuals



Batting_HBP Regression Summary

Formula: Q("Pitching_W-L%") ~ Q("Batting_HBP")

Coefficient Table

term	coef	std_err	t_stat	p_value	ci_lower	ci_upper
Intercept	0.3863	0.0544	7.0986	0.0000	0.2779	0.4946
Q("Batting_HBP")	0.0015	0.0007	2.1535	0.0344	0.0001	0.0029

Model Fit

	value
R-squared	0.0561
Adjusted R-squared	0.0440
F-statistic	4.6376
F-statistic p-value	0.0344
Observations	80.0000

Heteroskedasticity Tests

Test	LM Statistic	LM p-value	F Statistic	F p-value
Breusch-Pagan	2.7454	0.0975	2.7718	0.0999
White	2.9639	0.2272	1.4813	0.2338

Normality Test

JB statistic = 19.7107, JB p-value = 0.0001, Skewness = -0.7879, Kurtosis = 4.8520

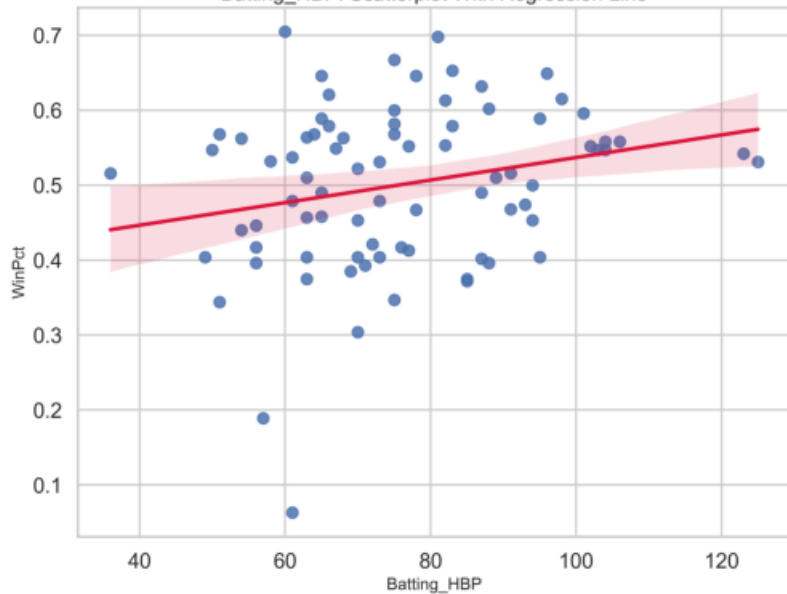
Interpretation

Batting_HBP has a statistically significant positive relationship with WinPct and explains approximately 5.6% of the variation in team winning percentage. Batting_HBP residuals do not exhibit significant heteroskedasticity under either the Breusch-Pagan or White tests. Residual normality is rejected at the 5% level.

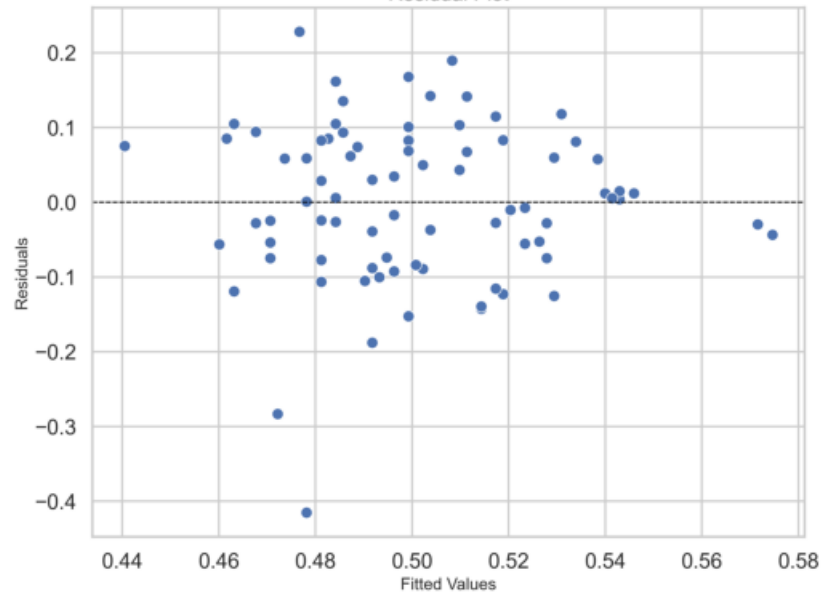
Warnings

- 4 observations exceed the leverage threshold of 0.0500.
- 3 observations exceed the Cook's distance threshold of 0.0500.

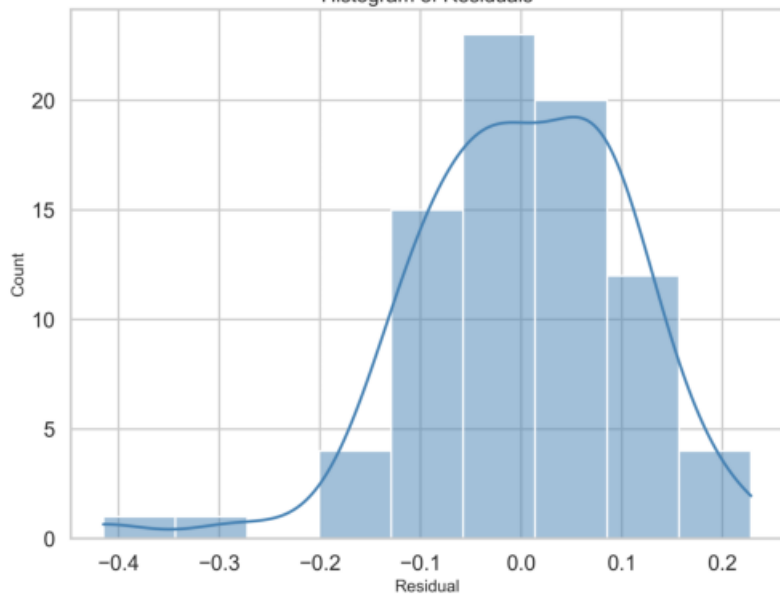
Batting_HBP: Scatterplot With Regression Line



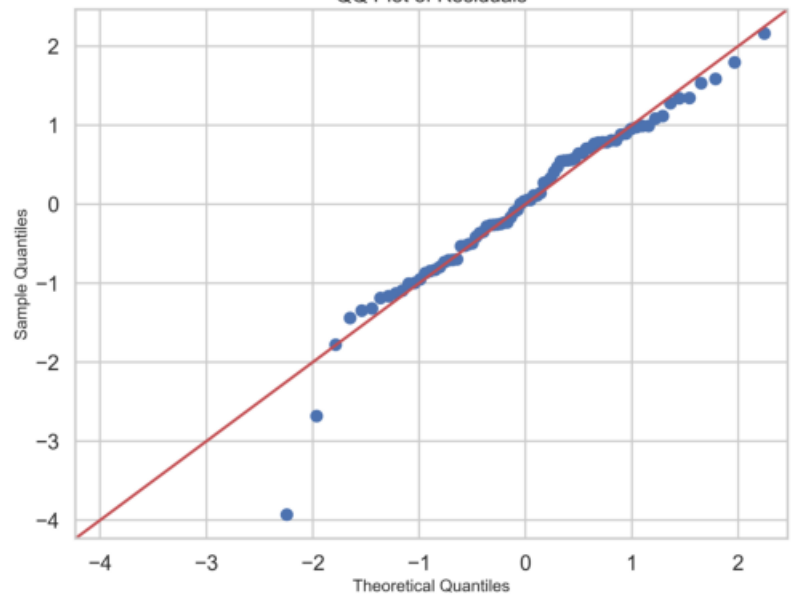
Residual Plot



Histogram of Residuals



QQ Plot of Residuals



Batting_SB Regression Summary

Formula: Q("Pitching_W-L%") ~ Q("Batting_SB")

Coefficient Table

term	coef	std_err	t_stat	p_value	ci_lower	ci_upper
Intercept	0.4359	0.0409	10.6468	0.0000	0.3544	0.5174
Q("Batting_SB")	0.0005	0.0003	1.6554	0.1019	-0.0001	0.0012

Model Fit

	value
R-squared	0.0339
Adjusted R-squared	0.0216
F-statistic	2.7403
F-statistic p-value	0.1019
Observations	80.0000

Heteroskedasticity Tests

Test	LM Statistic	LM p-value	F Statistic	F p-value
Breusch-Pagan	1.2839	0.2572	1.2722	0.2628
White	2.4801	0.2894	1.2317	0.2975

Normality Test

JB statistic = 18.7505, JB p-value = 0.0001, Skewness = -0.8641, Kurtosis = 4.6244

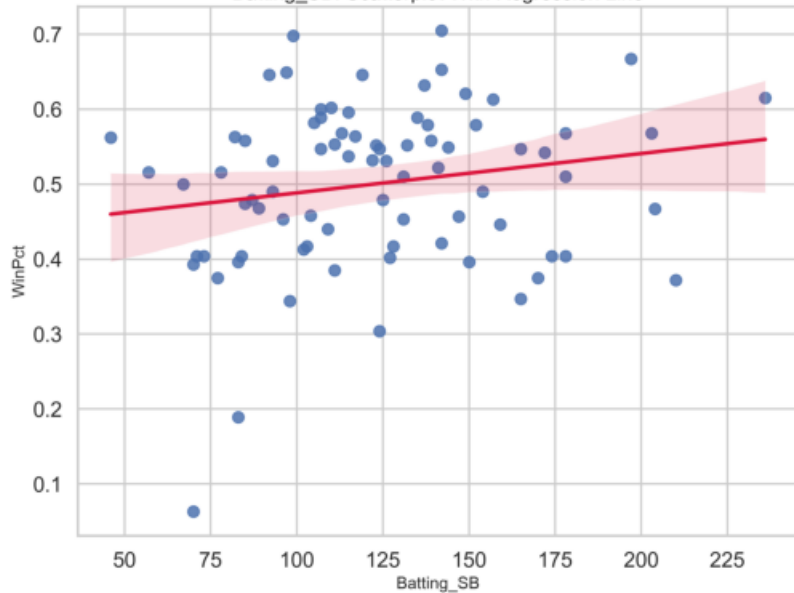
Interpretation

Batting_SB has a not statistically significant positive relationship with WinPct and explains approximately 3.4% of the variation in team winning percentage. Batting_SB residuals do not exhibit significant heteroskedasticity under either the Breusch-Pagan or White tests. Residual normality is rejected at the 5% level.

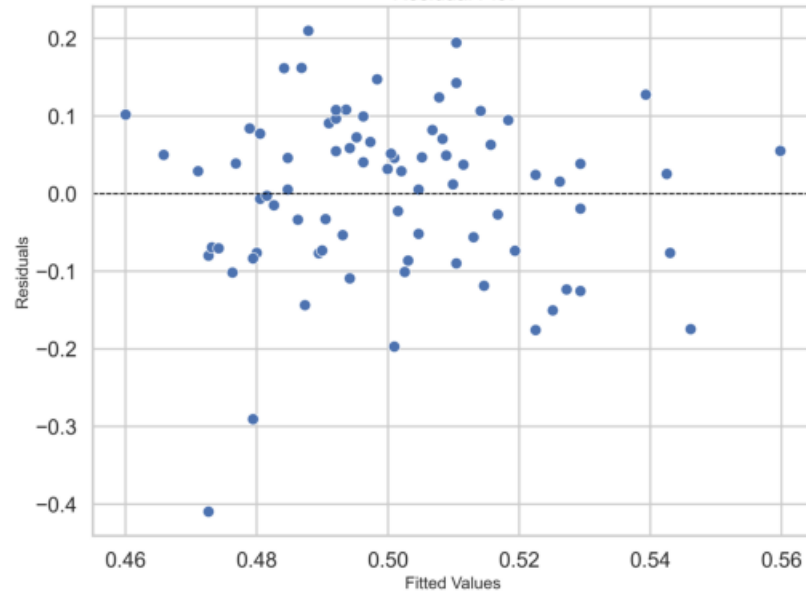
Warnings

- 7 observations exceed the leverage threshold of 0.0500.
- 3 observations exceed the Cook's distance threshold of 0.0500.

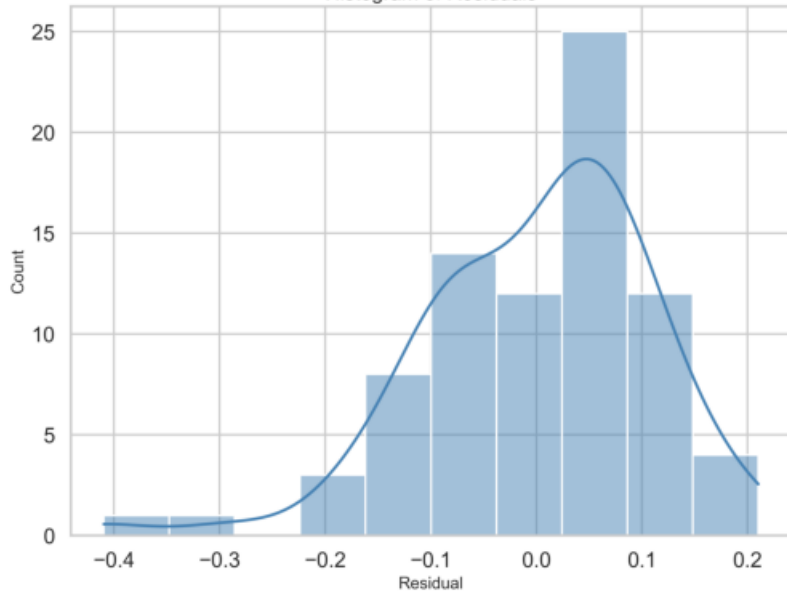
Batting_SB: Scatterplot With Regression Line



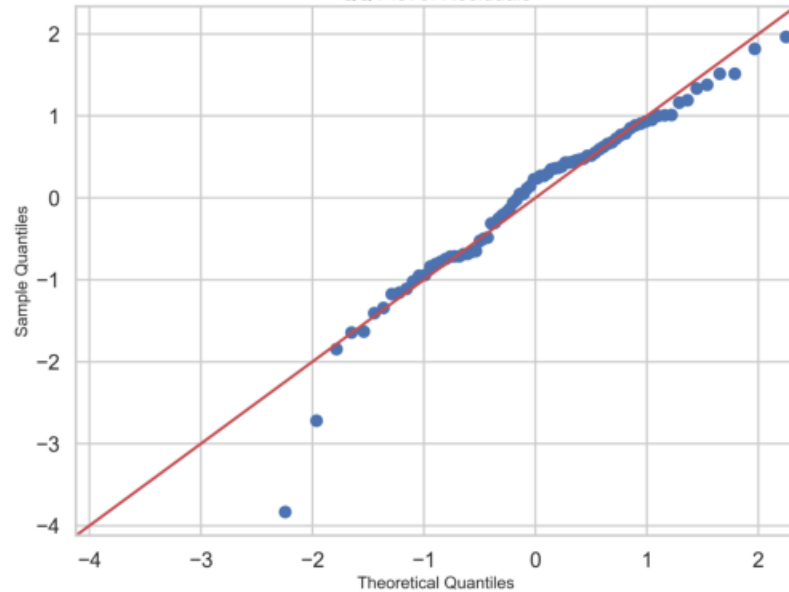
Residual Plot



Histogram of Residuals



QQ Plot of Residuals



Batting_G Regression Summary

Formula: $Q(\text{"Pitching_W-L\%"}) \sim Q(\text{"Batting_G"})$

Coefficient Table

term	coef	std_err	t_stat	p_value	ci_lower	ci_upper
Intercept	-1.2844	1.5444	-0.8316	0.4082	-4.3591	1.7904
Q("Batting_G")	0.0187	0.0162	1.1558	0.2513	-0.0135	0.0509

Model Fit

	value
R-squared	0.0168
Adjusted R-squared	0.0042
F-statistic	1.3358
F-statistic p-value	0.2513
Observations	80.0000

Heteroskedasticity Tests

Test	LM Statistic	LM p-value	F Statistic	F p-value
Breusch-Pagan	0.1092	0.7410	0.1067	0.7449
White	0.1363	0.9341	0.0657	0.9365

Normality Test

JB statistic = 35.3260, JB p-value = 0.0000, Skewness = -1.0066, Kurtosis = 5.5583

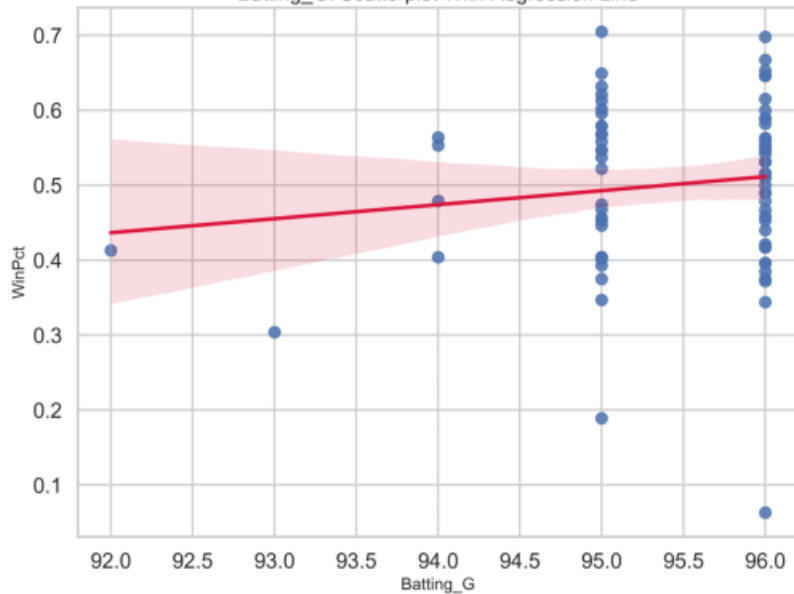
Interpretation

Batting_G has a not statistically significant positive relationship with WinPct and explains approximately 1.7% of the variation in team winning percentage. Batting_G residuals do not exhibit significant heteroskedasticity under either the Breusch-Pagan or White tests. Residual normality is rejected at the 5% level.

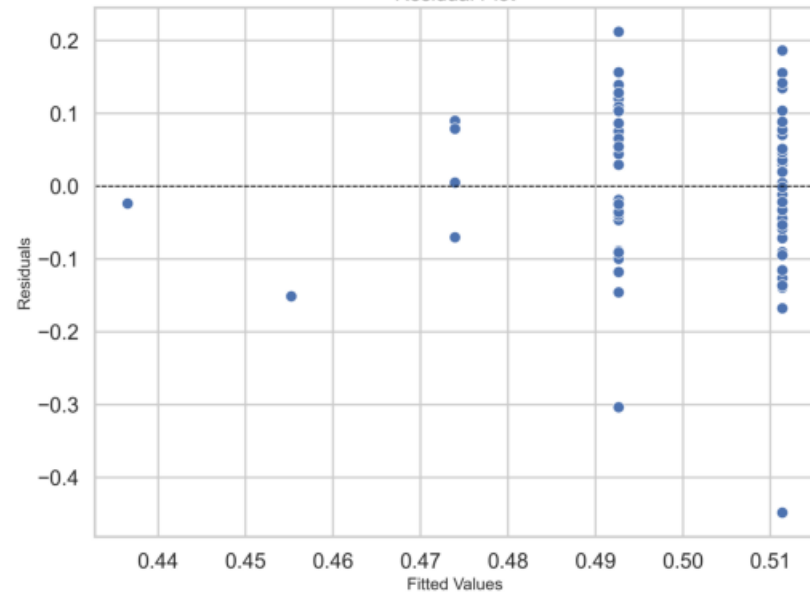
Warnings

6 observations exceed the leverage threshold of 0.0500.
3 observations exceed the Cook's distance threshold of 0.0500.

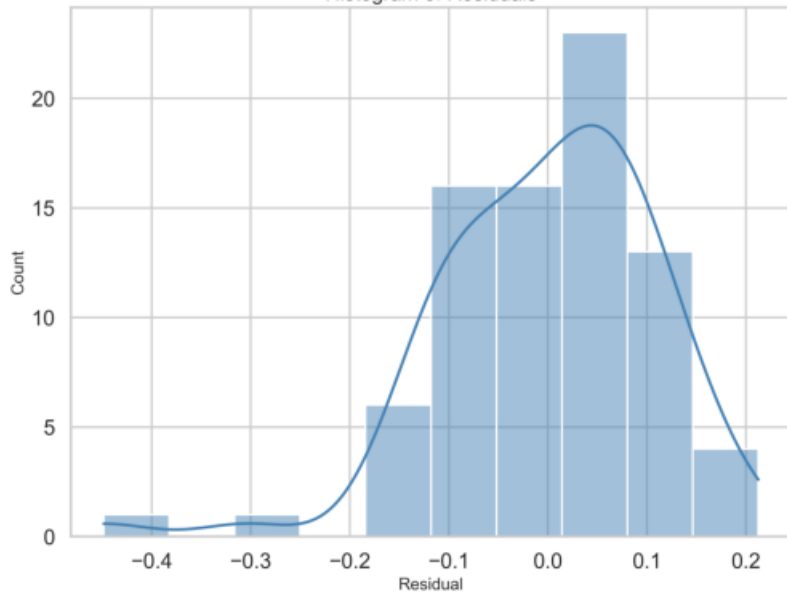
Batting_G: Scatterplot With Regression Line



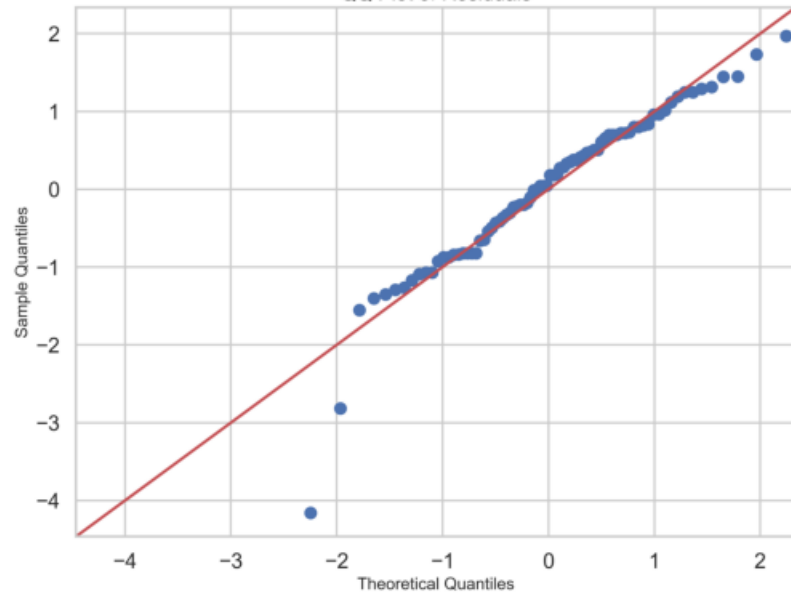
Residual Plot



Histogram of Residuals



QQ Plot of Residuals



Pitching_G Regression Summary

Formula: $Q(\text{"Pitching_W-L\%"}) \sim Q(\text{"Pitching_G"})$

Coefficient Table

	coef	std_err	t_stat	p_value	ci_lower	ci_upper
term						
Intercept	-1.2844	1.5444	-0.8316	0.4082	-4.3591	1.7904
Q("Pitching_G")	0.0187	0.0162	1.1558	0.2513	-0.0135	0.0509

Model Fit

	value
R-squared	0.0168
Adjusted R-squared	0.0042
F-statistic	1.3358
F-statistic p-value	0.2513
Observations	80.0000

Heteroskedasticity Tests

Test	LM Statistic	LM p-value	F Statistic	F p-value
Breusch-Pagan	0.1092	0.7410	0.1067	0.7449
White	0.1363	0.9341	0.0657	0.9365

Normality Test

JB statistic = 35.3260, JB p-value = 0.0000, Skewness = -1.0066, Kurtosis = 5.5583

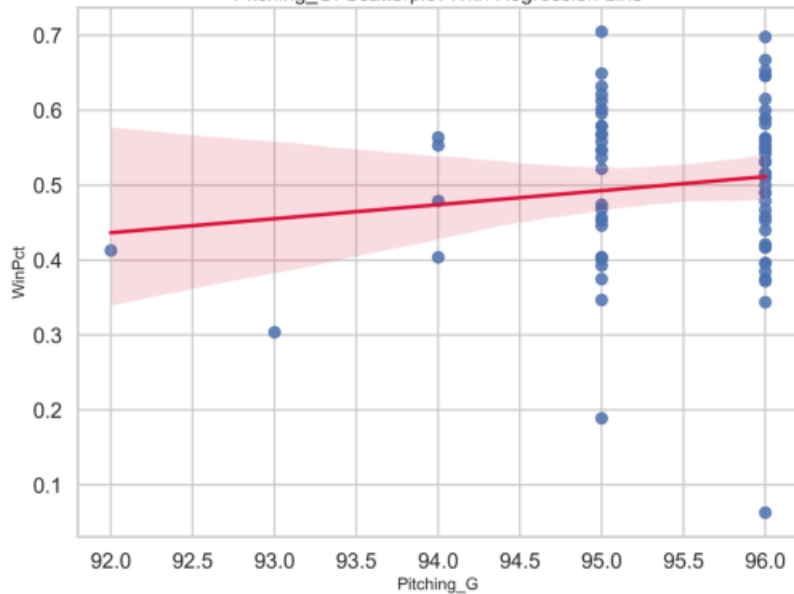
Interpretation

Pitching_G has a not statistically significant positive relationship with WinPct and explains approximately 1.7% of the variation in team winning percentage. Pitching_G residuals do not exhibit significant heteroskedasticity under either the Breusch-Pagan or White tests. Residual normality is rejected at the 5% level.

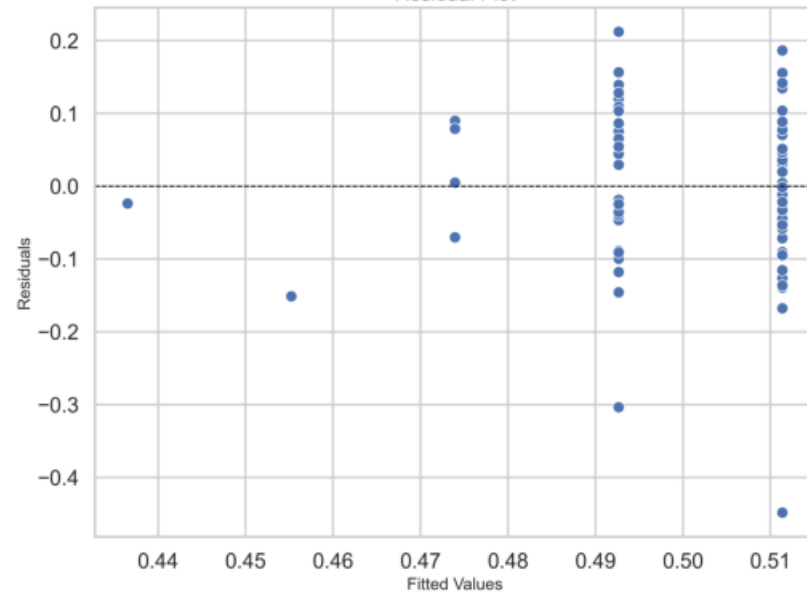
Warnings

6 observations exceed the leverage threshold of 0.0500.
3 observations exceed the Cook's distance threshold of 0.0500.

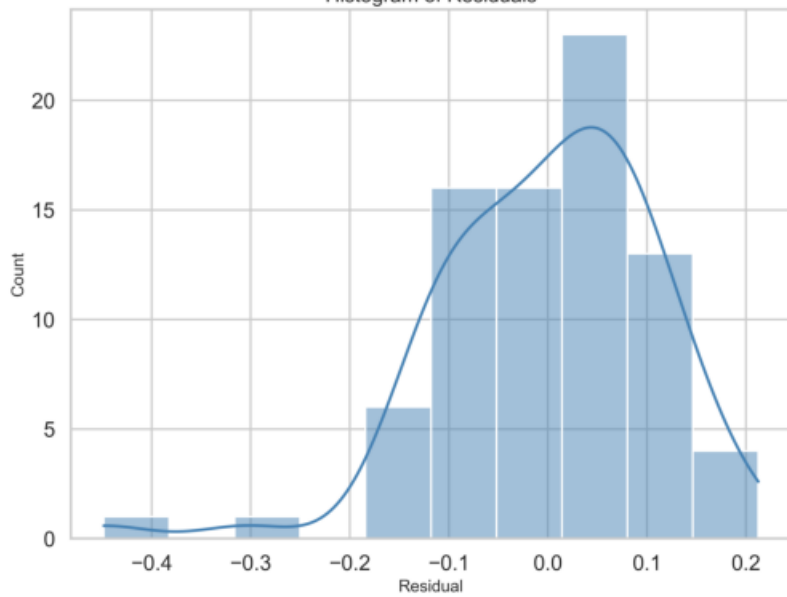
Pitching_G: Scatterplot With Regression Line



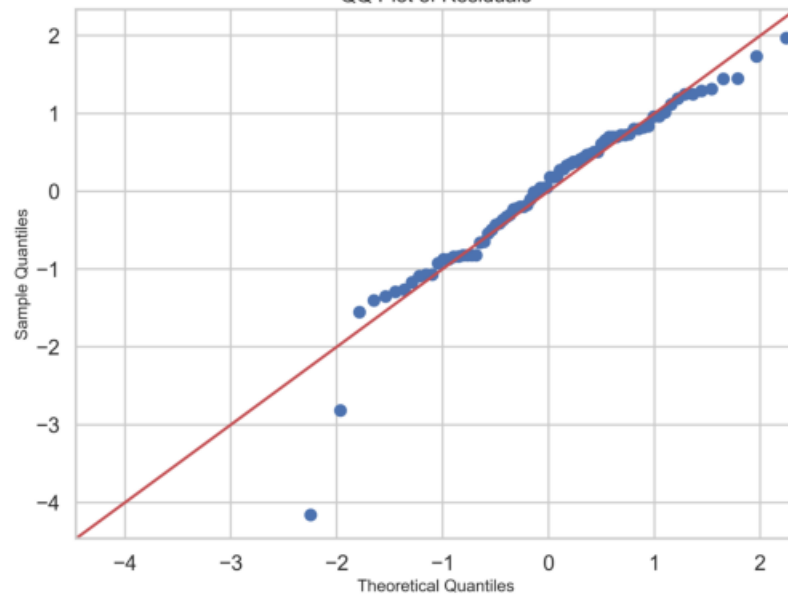
Residual Plot



Histogram of Residuals



QQ Plot of Residuals



Pitching_GS Regression Summary

Formula: $Q(\text{"Pitching_W-L\%"}) \sim Q(\text{"Pitching_GS"})$

Coefficient Table

	coef	std_err	t_stat	p_value	ci_lower	ci_upper
term						
Intercept	-0.8560	1.5592	-0.5490	0.5846	-3.9600	2.2481
Q("Pitching_GS")	0.0142	0.0163	0.8701	0.3869	-0.0183	0.0468

Model Fit

	value
R-squared	0.0096
Adjusted R-squared	-0.0031
F-statistic	0.7570
F-statistic p-value	0.3869
Observations	80.0000

Heteroskedasticity Tests

Test	LM Statistic	LM p-value	F Statistic	F p-value
Breusch-Pagan	0.1481	0.7004	0.1447	0.7047
White	0.1482	0.9286	0.0715	0.9311

Normality Test

JB statistic = 33.2161, JB p-value = 0.0000, Skewness = -0.9965, Kurtosis = 5.4480

Interpretation

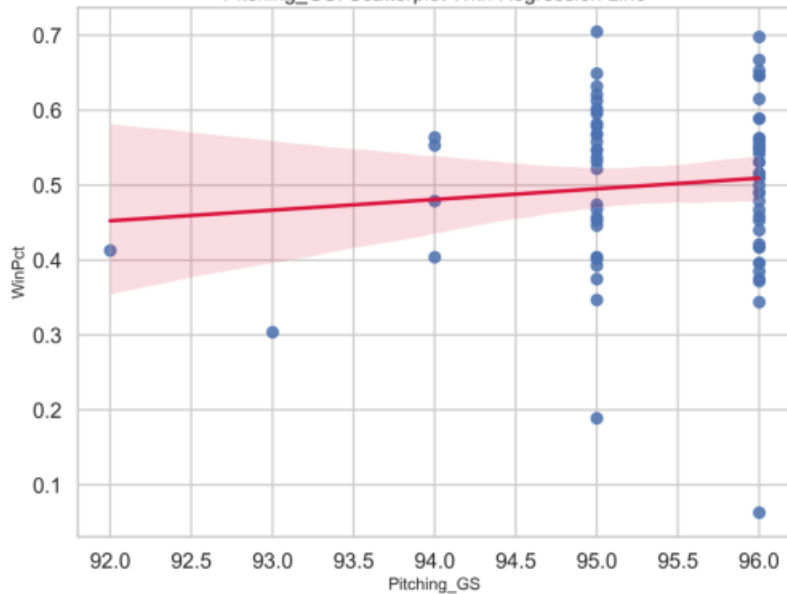
Pitching_GS has a not statistically significant positive relationship with WinPct and explains approximately 1.0% of the variation in team winning percentage. Pitching_GS residuals do not exhibit significant heteroskedasticity under either the Breusch-Pagan or White tests. Residual normality is rejected at the 5% level.

Warnings

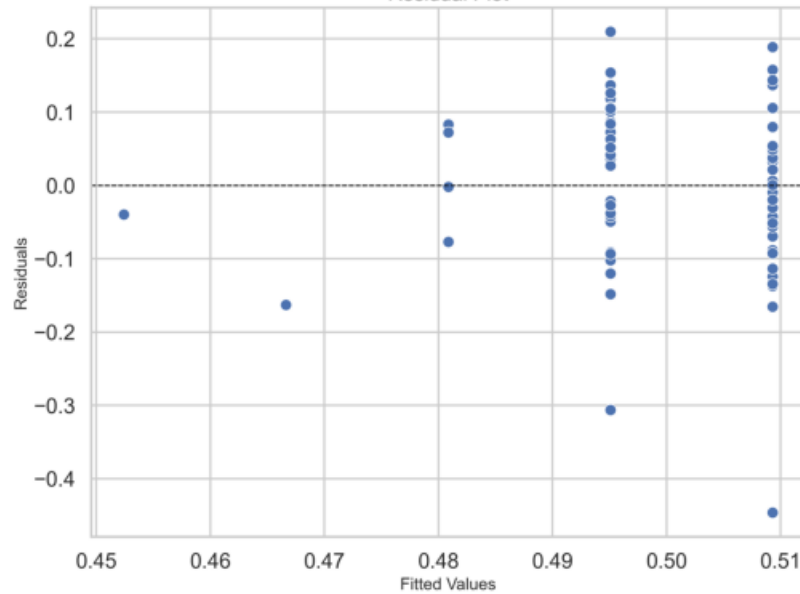
6 observations exceed the leverage threshold of 0.0500.

3 observations exceed the Cook's distance threshold of 0.0500.

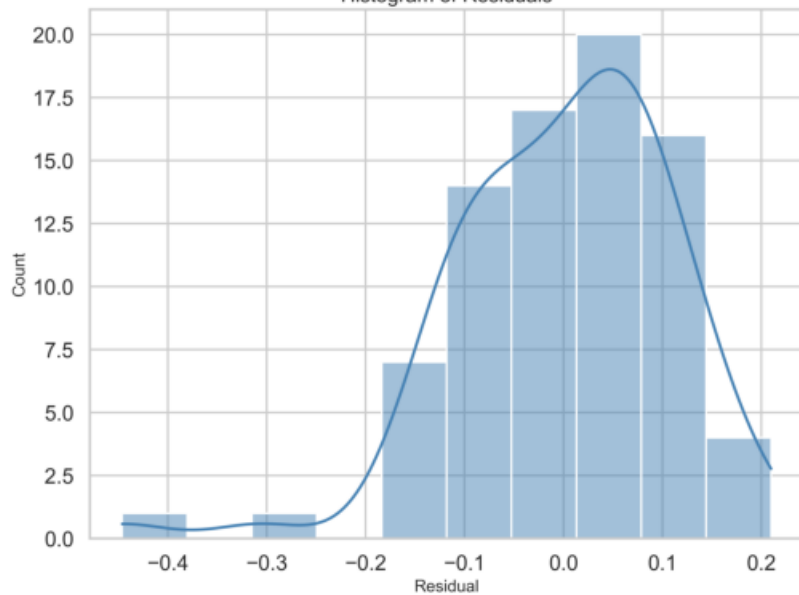
Pitching_GS: Scatterplot With Regression Line



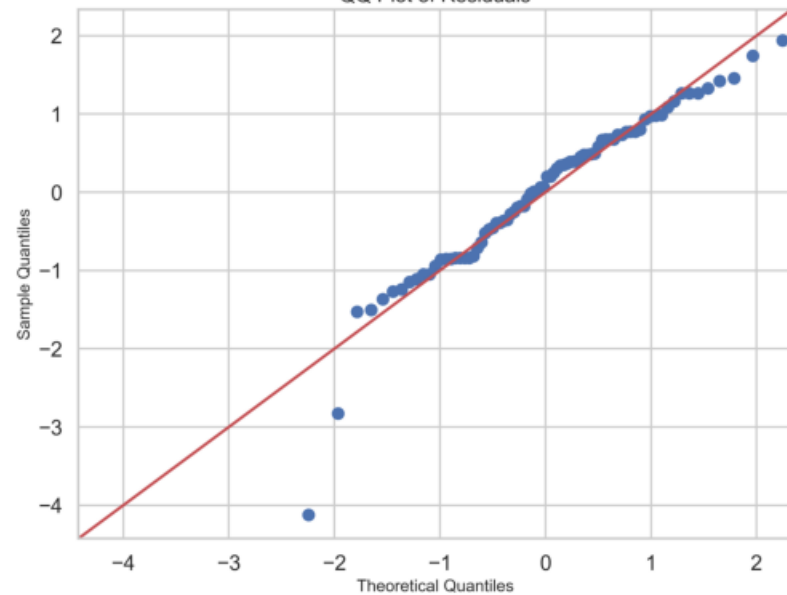
Residual Plot



Histogram of Residuals



QQ Plot of Residuals



Pitching_CG Regression Summary

Formula: Q("Pitching_W-L%") ~ Q("Pitching_CG")

Coefficient Table

	coef	std_err	t_stat	p_value	ci_lower	ci_upper
term						
Intercept	0.4859	0.0237	20.5271	0.0000	0.4388	0.5331
Q("Pitching_CG")	0.0049	0.0067	0.7232	0.4717	-0.0085	0.0182

Model Fit

	value
R-squared	0.0067
Adjusted R-squared	-0.0061
F-statistic	0.5230
F-statistic p-value	0.4717
Observations	80.0000

Heteroskedasticity Tests

Test	LM Statistic	LM p-value	F Statistic	F p-value
Breusch-Pagan	0.7451	0.3880	0.7333	0.3944
White	0.8124	0.6662	0.3950	0.6750

Normality Test

JB statistic = 27.5360, JB p-value = 0.0000, Skewness = -0.9279, Kurtosis = 5.1947

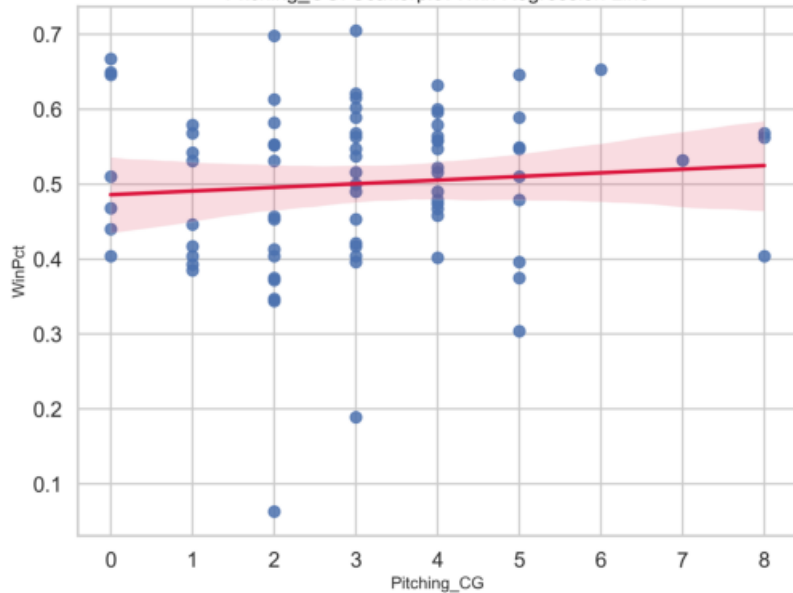
Interpretation

Pitching_CG has a not statistically significant positive relationship with WinPct and explains approximately 0.7% of the variation in team winning percentage. Pitching_CG residuals do not exhibit significant heteroskedasticity under either the Breusch-Pagan or White tests. Residual normality is rejected at the 5% level.

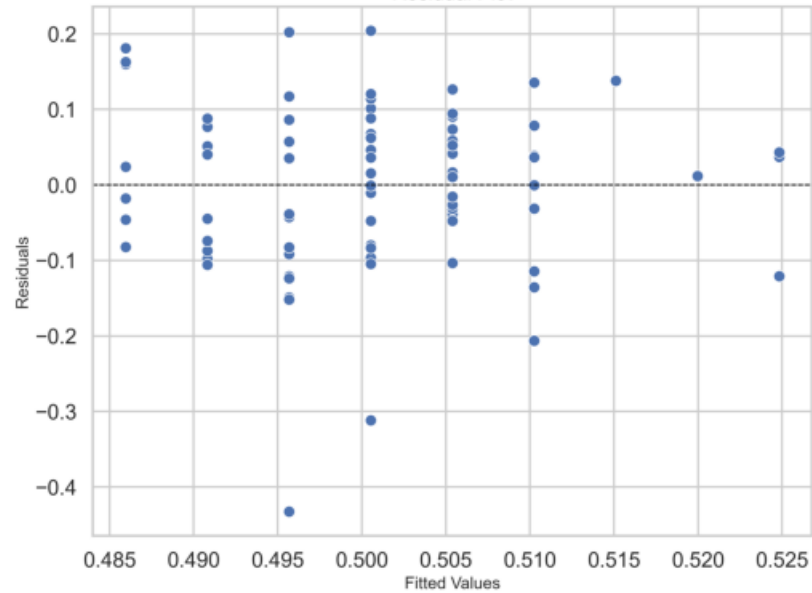
Warnings

- 4 observations exceed the leverage threshold of 0.0500.
- 7 observations exceed the Cook's distance threshold of 0.0500.

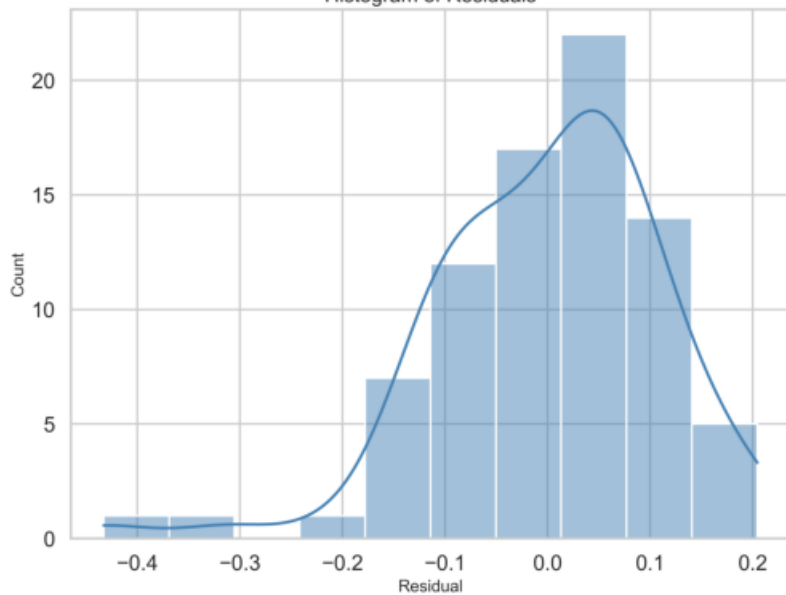
Pitching_CG: Scatterplot With Regression Line



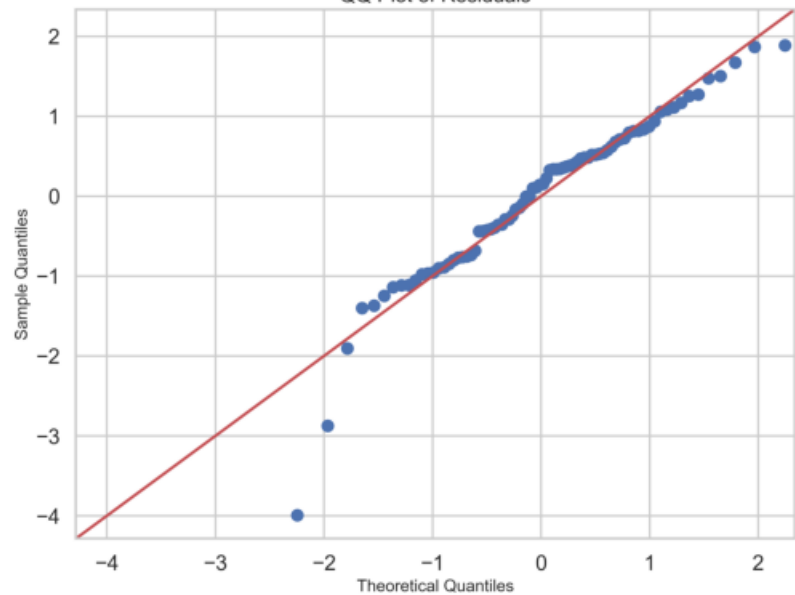
Residual Plot



Histogram of Residuals



QQ Plot of Residuals



Batting_3B Regression Summary

Formula: $Q(\text{"Pitching_W-L\%"}) \sim Q(\text{"Batting_3B"})$

Coefficient Table

term	coef	std_err	t_stat	p_value	ci_lower	ci_upper
Intercept	0.4821	0.0300	16.0832	0.0000	0.4225	0.5418
Q("Batting_3B")	0.0012	0.0017	0.6742	0.5021	-0.0023	0.0046

Model Fit

	value
R-squared	0.0058
Adjusted R-squared	-0.0070
F-statistic	0.4546
F-statistic p-value	0.5021
Observations	80.0000

Heteroskedasticity Tests

Test	LM Statistic	LM p-value	F Statistic	F p-value
Breusch-Pagan	0.0108	0.9174	0.0105	0.9186
White	0.5882	0.7452	0.2852	0.7527

Normality Test

JB statistic = 32.3747, JB p-value = 0.0000, Skewness = -0.9680, Kurtosis = 5.4422

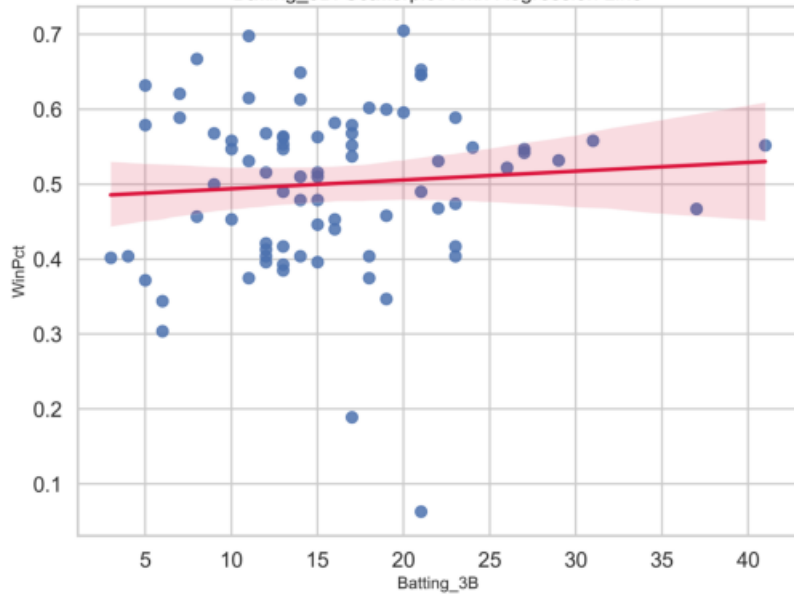
Interpretation

Batting_3B has a not statistically significant positive relationship with WinPct and explains approximately 0.6% of the variation in team winning percentage. Batting_3B residuals do not exhibit significant heteroskedasticity under either the Breusch-Pagan or White tests. Residual normality is rejected at the 5% level.

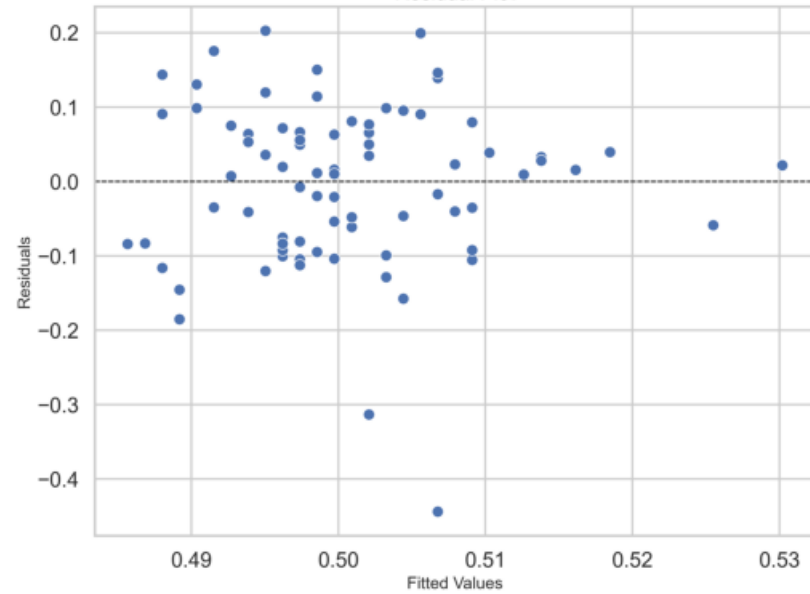
Warnings

5 observations exceed the leverage threshold of 0.0500.
3 observations exceed the Cook's distance threshold of 0.0500.

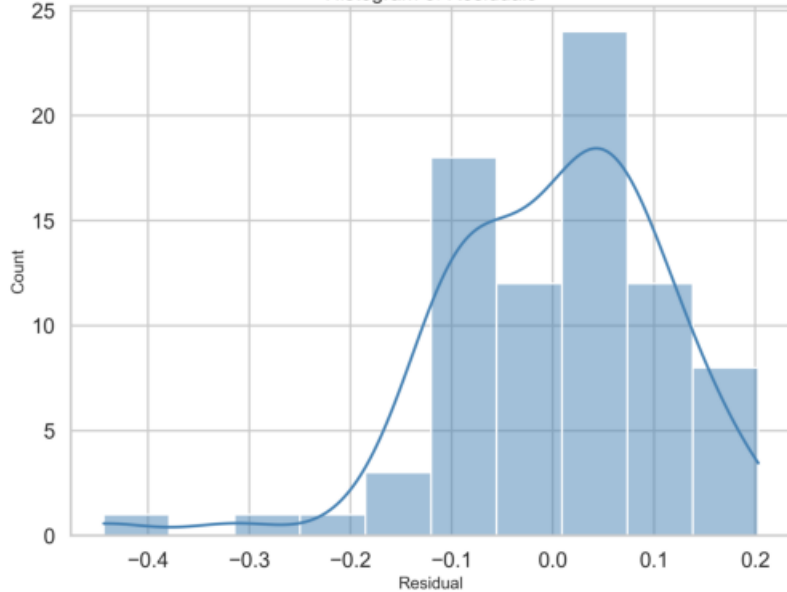
Batting_3B: Scatterplot With Regression Line



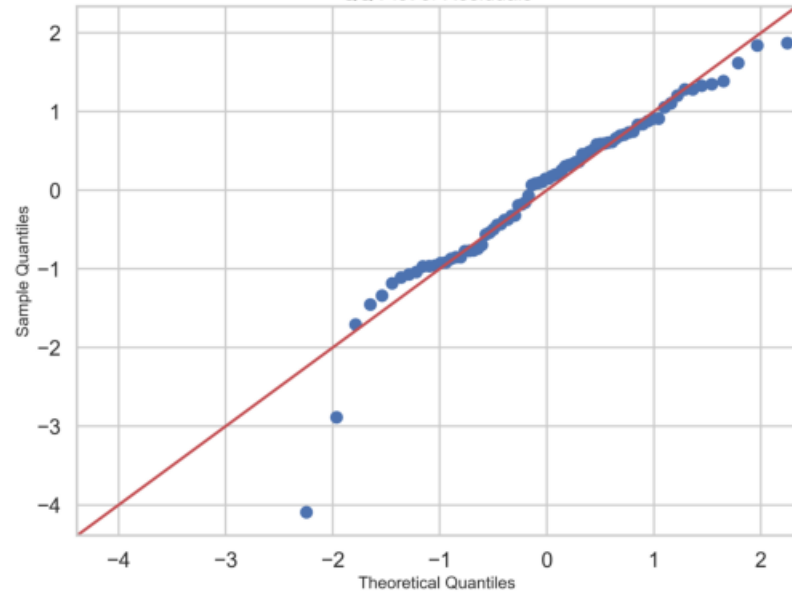
Residual Plot



Histogram of Residuals



QQ Plot of Residuals



Batting_CS Regression Summary

Formula: $Q(\text{"Pitching_W-L\%"}) \sim Q(\text{"Batting_CS"})$

Coefficient Table

term	coef	std_err	t_stat	p_value	ci_lower	ci_upper
Intercept	0.4794	0.0406	11.8091	0.0000	0.3986	0.5602
$Q(\text{"Batting_CS"})$	0.0006	0.0012	0.5473	0.5857	-0.0017	0.0029

Model Fit

	value
R-squared	0.0038
Adjusted R-squared	-0.0089
F-statistic	0.2996
F-statistic p-value	0.5857
Observations	80.0000

Heteroskedasticity Tests

Test	LM Statistic	LM p-value	F Statistic	F p-value
Breusch-Pagan	0.0408	0.8399	0.0398	0.8424
White	0.2326	0.8902	0.1123	0.8939

Normality Test

JB statistic = 26.6174, JB p-value = 0.0000, Skewness = -0.9618, Kurtosis = 5.0700

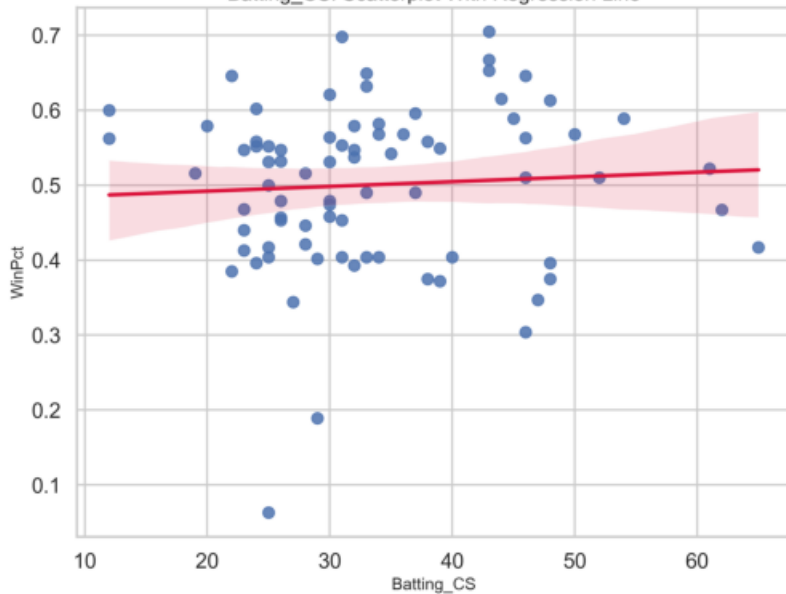
Interpretation

Batting_CS has a not statistically significant positive relationship with WinPct and explains approximately 0.4% of the variation in team winning percentage. Batting_CS residuals do not exhibit significant heteroskedasticity under either the Breusch-Pagan or White tests. Residual normality is rejected at the 5% level.

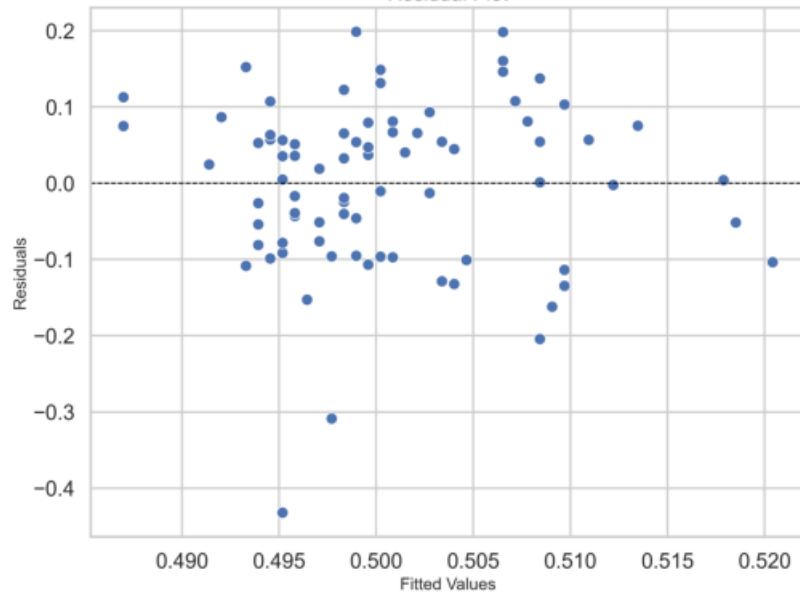
Warnings

6 observations exceed the leverage threshold of 0.0500.
4 observations exceed the Cook's distance threshold of 0.0500.

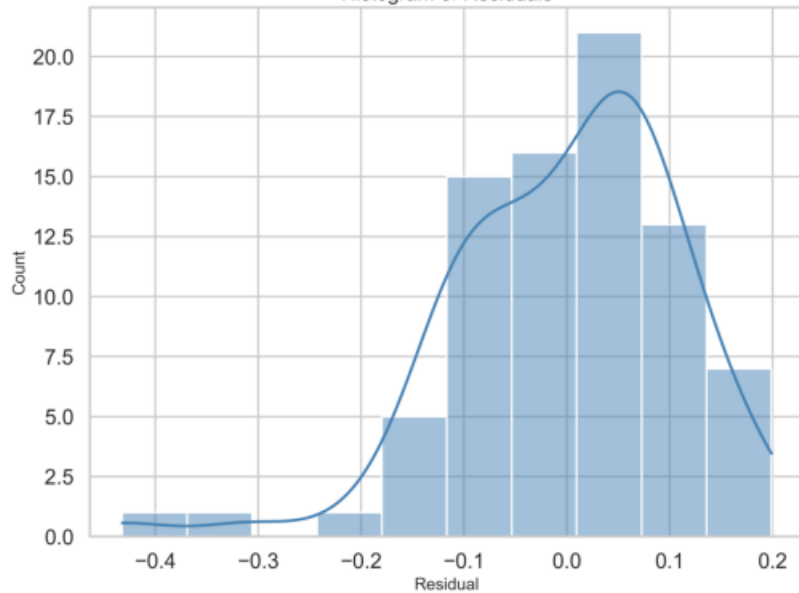
Batting_CS: Scatterplot With Regression Line



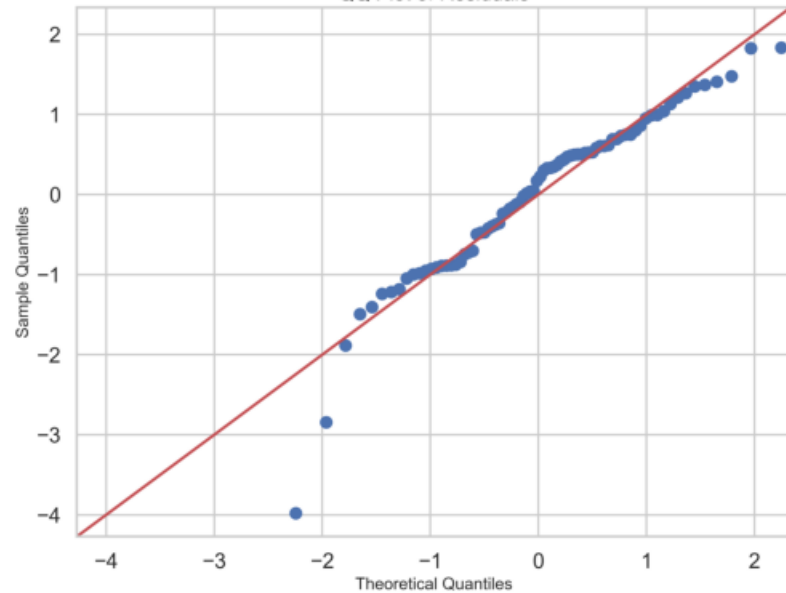
Residual Plot



Histogram of Residuals



QQ Plot of Residuals



Batting_SH Regression Summary

Formula: $Q(\text{"Pitching_W-L\%"}) \sim Q(\text{"Batting_SH"})$

Coefficient Table

term	coef	std_err	t_stat	p_value	ci_lower	ci_upper
Intercept	0.4904	0.0293	16.7508	0.0000	0.4321	0.5487
$Q(\text{"Batting_SH"})$	0.0005	0.0012	0.3831	0.7027	-0.0019	0.0028

Model Fit

	value
R-squared	0.0019
Adjusted R-squared	-0.0109
F-statistic	0.1468
F-statistic p-value	0.7027
Observations	80.0000

Heteroskedasticity Tests

Test	LM Statistic	LM p-value	F Statistic	F p-value
Breusch-Pagan	0.0373	0.8468	0.0364	0.8492
White	0.1911	0.9089	0.0922	0.9120

Normality Test

JB statistic = 28.8139, JB p-value = 0.0000, Skewness = -0.9625, Kurtosis = 5.2223

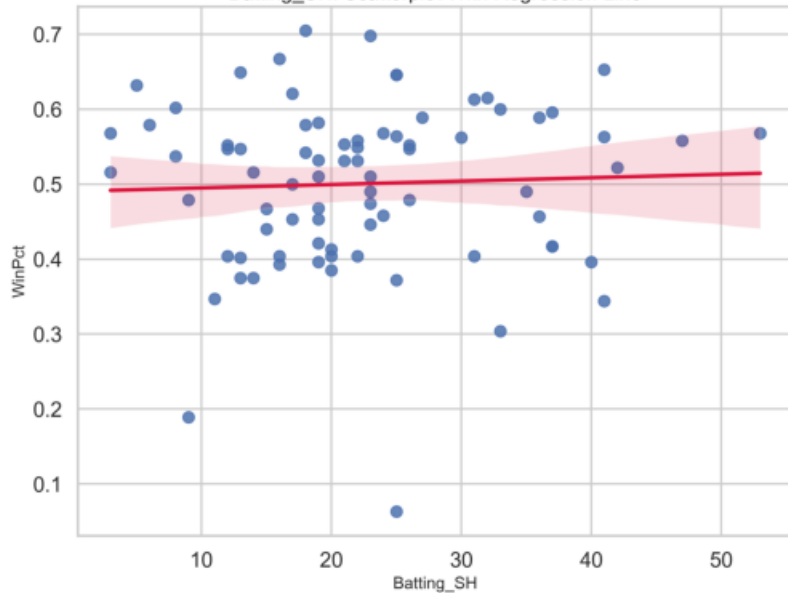
Interpretation

Batting_SH has a not statistically significant positive relationship with WinPct and explains approximately 0.2% of the variation in team winning percentage. Batting_SH residuals do not exhibit significant heteroskedasticity under either the Breusch-Pagan or White tests. Residual normality is rejected at the 5% level.

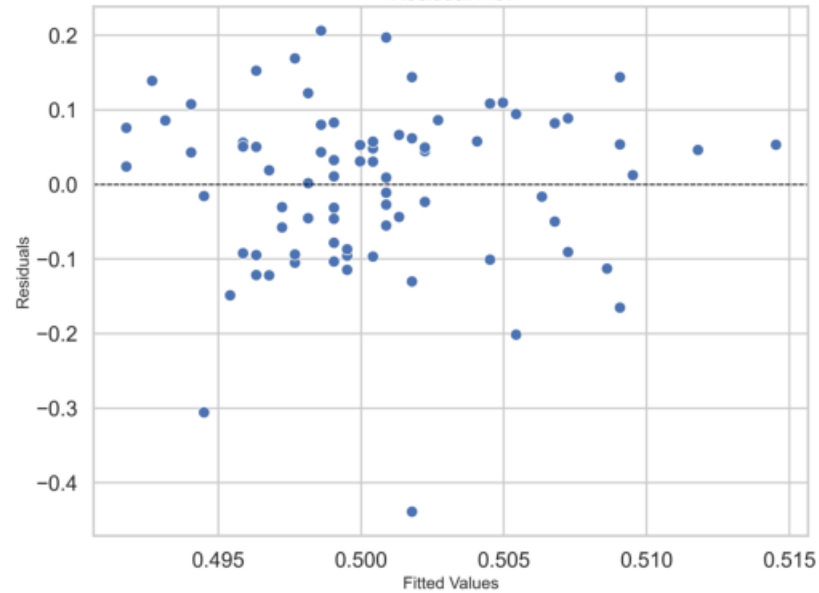
Warnings

8 observations exceed the leverage threshold of 0.0500.
4 observations exceed the Cook's distance threshold of 0.0500.

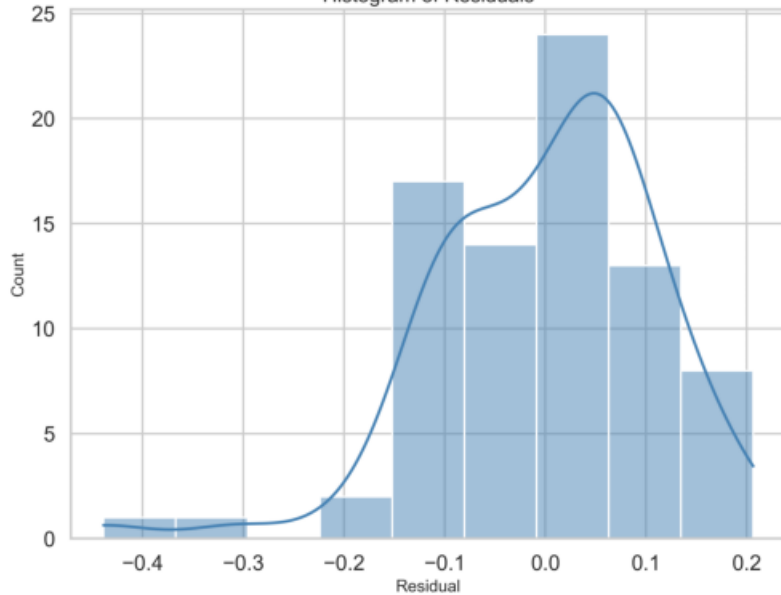
Batting_SH: Scatterplot With Regression Line



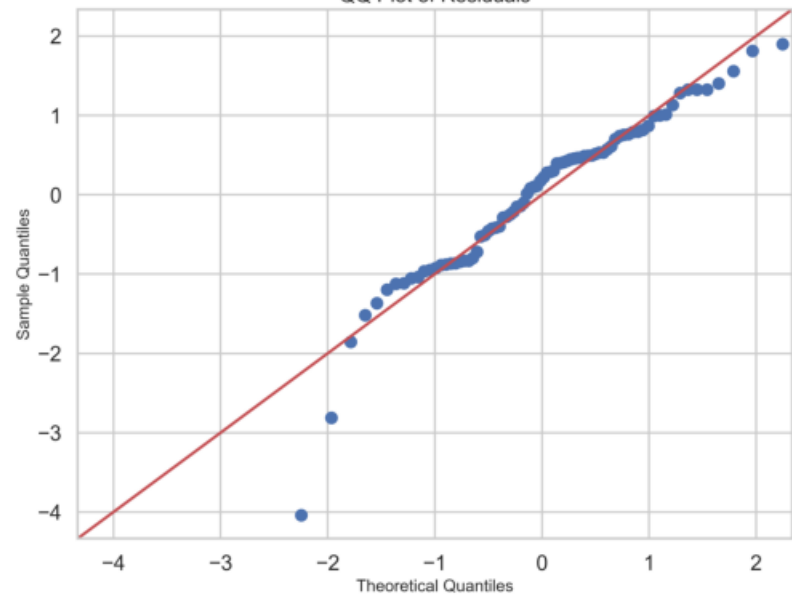
Residual Plot



Histogram of Residuals



QQ Plot of Residuals



Pitching_GF Regression Summary

Formula: $Q(\text{"Pitching_W-L\%"}) \sim Q(\text{"Pitching_GF"})$

Coefficient Table

	coef	std_err	t_stat	p_value	ci_lower	ci_upper
term						
Intercept	0.6876	0.5644	1.2183	0.2268	-0.4360	1.8113
Q("Pitching_GF")	-0.0020	0.0061	-0.3315	0.7412	-0.0142	0.0101

Model Fit

	value
R-squared	0.0014
Adjusted R-squared	-0.0114
F-statistic	0.1099
F-statistic p-value	0.7412
Observations	80.0000

Heteroskedasticity Tests

Test	LM Statistic	LM p-value	F Statistic	F p-value
Breusch-Pagan	0.5461	0.4599	0.5361	0.4662
White	0.6520	0.7218	0.3164	0.7297

Normality Test

JB statistic = 27.7081, JB p-value = 0.0000, Skewness = -0.9478, Kurtosis = 5.1724

Interpretation

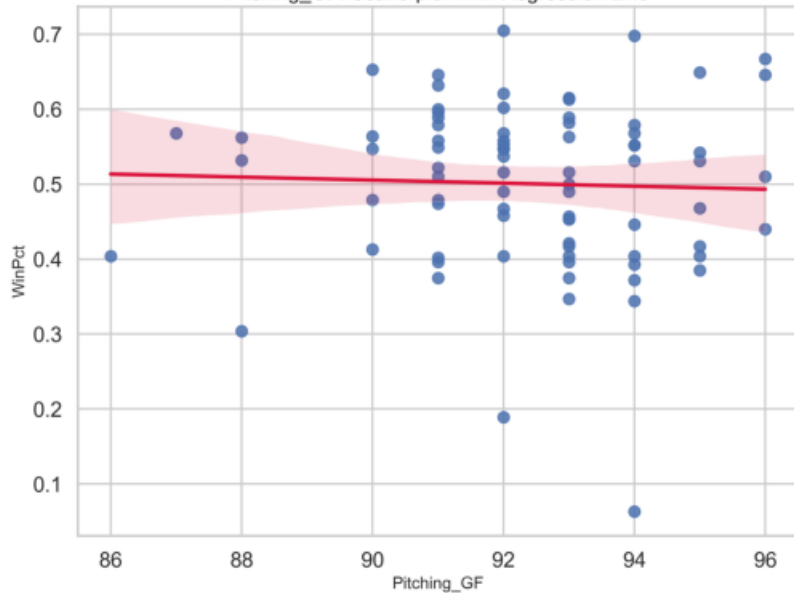
Pitching_GF has a not statistically significant negative relationship with WinPct and explains approximately 0.1% of the variation in team winning percentage. Pitching_GF residuals do not exhibit significant heteroskedasticity under either the Breusch-Pagan or White tests. Residual normality is rejected at the 5% level.

Warnings

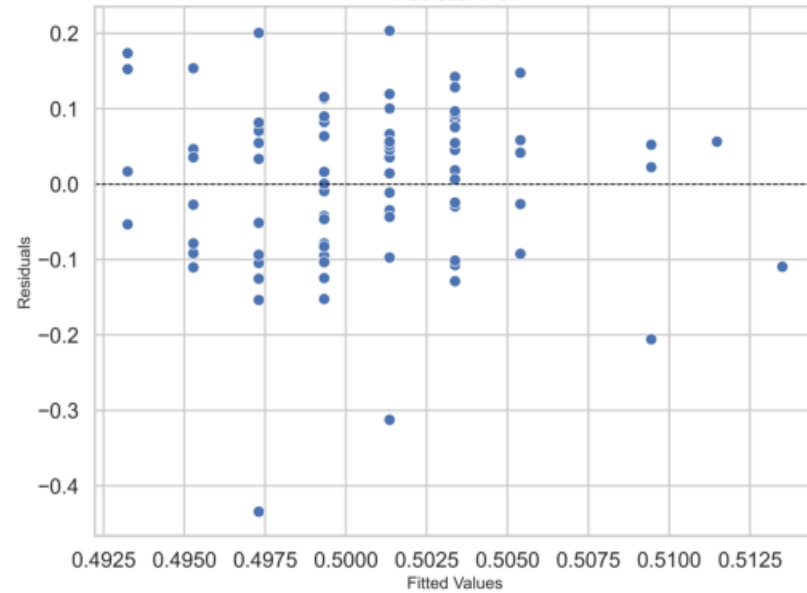
9 observations exceed the leverage threshold of 0.0500.

6 observations exceed the Cook's distance threshold of 0.0500.

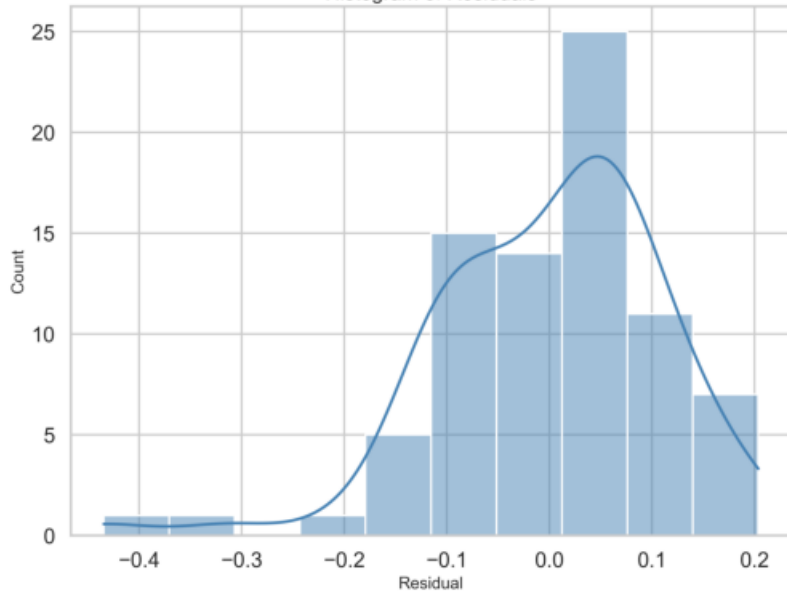
Pitching_GF: Scatterplot With Regression Line



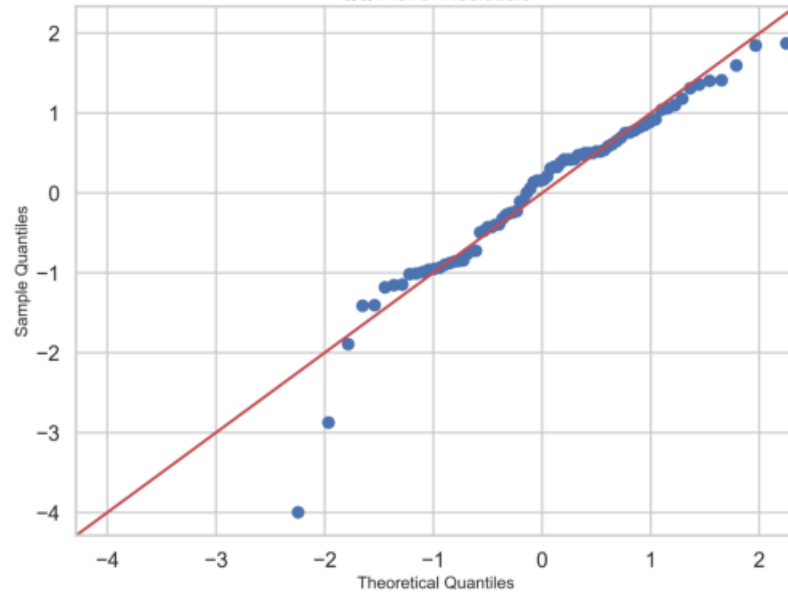
Residual Plot



Histogram of Residuals



QQ Plot of Residuals



Final Batch Findings

Best predictor of WinPct: RA9 (R-squared = 0.5973)

Cleanest residual behavior: ERA (assumption score = 0.7940)

Most severe assumption violations: Pitching_GF (assumption score = 2.2174)